

PROJECT

BANKING AND FINANCE DOMAIN

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BANKING AND FINANCE DOMAIN

Go to your repository and create a new repository in the name of “*project-1*” → Now download the project into zip file using the following link

“<https://github.com/StarAgileDevOpsTraining/star-agile-banking-finance.git>” → Now extract the downloaded folder and paste the folder in the desktop → Folder name will be “*star-agile-banking-finance-master*” move the folder to desktop and rename it as “*Project-1*” → Now install the git bash → Now go inside the folder and right click and open git bash terminal → Now give the command “*git init*” to initialize the git in the current folder → Now add remote github using the command “*git remote add origin <https://github.com/Balasastha/Project-1>*” → Create a PAT personal access token in the github → Go to github profile → settings → Developer settings → Personal Access tokens and generate the token → First add all the files using the command “*git add .*” and make commit using the command “*git commit -m "Initial commit"*” Now push the file from git to github using the command “*git push origin -u master*” → It asks for PAT just copy and paste it from github

and sign in → This will push file to github → Now create 3 EC2 ubuntu instance with all traffic enabled using terraform → First download terraform from the *“terraform.io”* website and create a folder in the desktop named as *“terraform-scripts”* now go to the downloaded folder and extract it and copy the application file and paste it in the *“terraform-scripts”* folder → Now download Visual Studio Code → Open the folder that we have created in desktop → Inside the folder create a file names as *“main.tf”* and type the following content inside it,

```
provider "aws" {
```

```
    region = "ap-south-1" # India region
```

```
}
```

```
resource "aws_security_group" "allow_all" {
```

```
    name      = "allow_all"
```

```
    description = "Allow all inbound and outbound traffic"
```

```
    ingress {
```

```
        from_port = 0
```

```
        to_port   = 65535
```

```
        protocol  = "tcp"
```

```
cidr_blocks = ["0.0.0.0/0"]

}

egress {

from_port = 0

to_port = 65535

protocol = "tcp"

cidr_blocks = ["0.0.0.0/0"]

}

}

resource "aws_instance" "master_server" {

ami = "ami-053b12d3152c0cc71" # Ubuntu AMI ID

instance_type = "t2.medium" # EC2 instance type

security_groups = [aws_security_group.allow_all.name]

key_name = "Project-1" # SSH key name

root_block_device {

volume_size = 15 # 15 GB storage
```



```
volume_type = "gp2" # General Purpose SSD

}

tags = {

    Name = "Master-server"

}

}

resource "aws_instance" "grafana_server" {

    ami          = "ami-053b12d3152c0cc71" # Ubuntu AMI ID

    instance_type = "t2.medium"

    security_groups = [aws_security_group.allow_all.name]

    key_name       = "Project-1"

    root_block_device {

        volume_size = 15

        volume_type = "gp2"

    }
```

```
tags = {  
  
    Name = "Grafana"  
  
}  
  
}
```

```
resource "aws_instance" "test_server" {  
  
    ami          = "ami-053b12d3152c0cc71" # Ubuntu AMI ID  
  
    instance_type = "t2.medium"  
  
    security_groups = [aws_security_group.allow_all.name]  
  
    key_name      = "Project-1"
```

```
root_block_device {  
  
    volume_size = 15  
  
    volume_type = "gp2"  
  
}
```

```
tags = {  
  
    Name = "Test-server"  
  
}
```

}

Now to access the aws account we have open the terminal in VS Code using the cmd “*ctrl + `*” and type “*aws configure*” there it will ask for access key and secret access key which will be available under profile section → Security credentials in aws account → There you can create keys and copy and paste it in the VS terminal → Now type the command “*terraform init*” to initialize and type the command “*terraform plan*” and type the command “*terraform apply*” and enter the value as “*yes*” this will create the instance accordingly → Now go to the aws account and check for the instance the instances will be created in which one instance is “*Master-server*” and another machine is “*grafana*” and another one is “*Test-server*” → Now connect the three machines → Go inside the master machine and type the following command “*sudo su*” to gain root privilege access and update all packages using the command “*apt update*” → Now install java using the following command “*sudo apt install openjdk-11-jdk -y*” → Now install Jenkins in it using the following commands,

“*wget <https://raw.githubusercontent.com/akshu20791/Deployment-script/main/jenkins.sh>*”

“*chmod +x jenkins.sh*”

“*./jenkins.sh*”

Now copy the public ip of the machine and paste it in the browser like **"<public-ip>:8080"** → It will ask for administrator password which is in the machine copy and paste it in the Jenkins → Install suggested plugins → Install docker using the command **"sudo apt install docker.io -y"** → To add the jenkins user to the docker group use the command **"sudo usermod -aG docker jenkins"** → Now give root user access to jenkins for go to sudoers file by using the cmd **"visudo"** → Under user privilege specification add **"jenkins ALL=(ALL:ALL) NOPASSWD: ALL"** → Press **"ctrl + x"** and **"y"** to save and exit → Restart the jenkins using command **"service jenkins restart"** → Go to jenkins and refresh it and login again → Create new item → Give the project name and choose pipeline → Go to pipeline syntax to add docker credentials → Choose sample step as **"withCredentials: Bind Credentials to variables"** → Under bindings click **"Add"** → Secret text → Add → Jenkins → Under kind choose **"secret text"** → In secret file type the password of your docker account → In ID type as **"dockerhubpass"** → Under description type **"To hide docker hub password"** → Click on **"Generate Pipeline Script"** → Just copy the pipeline that is generated which will be used in jenkins pipeline to login into docker → Go to the master machine and install ansible using following commands,

"sudo apt install software-properties-common"

"sudo add-apt-repository --yes --update ppa:ansible/ansible"

“sudo apt install ansible”

Go to jenkins → Manage jenkins → Plugins → Available plugins → Search for ***“docker”*** plugin and install it and ***“ansible”*** plugin and install it → Go to manage jenkins → Tools → Under ansible installation ***“Add ansible”*** → Name it as ***“ansible”*** → Check the box ***“Install automatically”*** → Apply and save → Now go to master machine and type the command ***“vi /etc/ansible/hosts”*** → In that file type the following,

“[ansiblegroup]

Private IP of test server”

Now configure the ansible file in the local repo and push it into remote repo using the command ***“vi ansible-playbook.yml”*** → Inside type the following content,

“- name : Configure Docker on EC2 Instances

hosts : all

become: true

connection : ssh

tasks :

- name: updating apt

command : sudo apt-get update

- **name** : Install Docker

command : `sudo apt-get install -y docker.io`

- **name** : Start Docker Service

command : `sudo systemctl start docker`

- **name**: Deploy Docker Container

command: `docker run -itd -p 8084:8081 balasastha/banking-finance:1`

Save and exit the file now add the file using the command “***git add ansible-playbook.yml***” → Now commit the changes using command “***git commit -m “Second commit”***” → Now push the command to the github repo “***git push origin -u master***” → Now go to jenkins → Go to the project → Pipeline → Pipeline syntax → Under Sample step choose “***ansiblePlaybook: Invoke an ansible playbook***” → Under ansible tool choose “***ansible***” → Specify playbook file path in workspace as “***ansible-playbook.yml***” → Specify inventory file path in workspace as “***/etc/ansible/hosts***” → Under SSH connection credentials → Click “***Add***” → Jenkins → Under kind choose “***SSH username with private key***” → Specify id as “***ansible***” → Specify description “***Private key file***”

for test server connection” → Specify username as “*ubuntu*” → Check “*Enter directly*” option and paste the private key file → Under SSH connection credentials select “*ubuntu (Private key for test server connection)*” → Check the option “*Use become*” → Check “*Disable the host SSH key check*” → Click on “*Generate pipeline*” → Just copy the pipeline that is generated which will be used in jenkins pipeline to establish connection to the test server → Now create a jenkins file in the local repo and push it into remote repo using the command “*vi Jenkinsfile*” → Paste the following content inside the file,

```
pipeline {
```

```
    agent any
```

```
    stages{
```

```
        stage('Git checkout'){
```

```
            steps
```

```
            {
```

```
                echo 'Checking out the git repo...'
```

```
                git url:' https://github.com/Balasastha/Project-1.git'
```

```
            }
```

```
        }
```

```
        stage('Package the code'){
```

steps

{

echo 'Packaging the code for you...'

sh 'mvn clean package'

}

}

stage ('Build docker image'){

steps

{

echo 'Building docker image for you..'

sh 'docker build -t balasastha/banking-finance:1 .'

}

}

stage ('Docker login'){

steps

{

echo 'Logging in into docker...'

withCredentials([string(credentialsId: 'dockerhubpass',

variable: 'dockerhubpass')) {


```
sh 'docker login -u balasastha -p ${dockerhubpass}'
```

```
}
```

```
}
```

```
}
```

```
stage ('Push image into docker'){
```

```
steps
```

```
{
```

```
echo 'Pushing the docker image for you'
```

```
sh 'docker push balasastha/banking-finance:1'
```

```
}
```

```
}
```

```
stage ('Invoking ansible playbook'){
```

```
steps
```

```
{
```

```
echo 'Running ansible playbook in test server...'
```

```
ansiblePlaybook become: true, credentialsId: 'ansible',
```

```
disableHostKeyChecking: true, installation: 'ansible', inventory:
```

```
'/etc/ansible/hosts', playbook: 'ansible-playbook.yml', vaultTmpPath:
```

```
"
```

```
}  
  
}  
  
}  
  
}
```

Save and quit the file → Add file using the command “***git add Jenkinsfile***” → Now commit the changes using the command “***git commit -m “Third commit”***” → Now push the Jenkins file to the github repo using the command “***git push origin -u master***” → Go to the jenkins and go to the pipeline of the created project → Under definition choose “***Pipeline script from SCM***” → In SCM choose “***Git***” → In repository URL give the link of the repo that we are using “**<https://github.com/Balasastha/Project-1>**” → Now build the project by clicking the option “***Build now***” → The project will be build → Copy the public IP on the test machine and put it onto the browser “***<Public IP of the machine>:8084***” in that application will be running → Now setup Prometheus and grafana in grafana machine → For that you have to install node exporter in both master and test server → To install node exporter connect the 2 machine master and test server and create a .sh file using the command “***vi install_node_exporter.sh***” and type the following command,

#!/bin/bash

Exit the script if any command fails

set -e

echo "Starting Node Exporter installation..."

Step 1: Download Node Exporter

echo "Downloading Node Exporter..."

wget

https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz -O /tmp/node_exporter.tar.gz

Step 2: Extract the archive

echo "Extracting Node Exporter..."

tar -xvzf /tmp/node_exporter.tar.gz -C /tmp/

Step 3: Move the binary

echo "Moving Node Exporter binary to /usr/local/bin..."

*sudo mv /tmp/node_exporter-1.6.1.linux-amd64/node_exporter
/usr/local/bin/*

Step 4: Create a systemd service file

echo "Creating systemd service file for Node Exporter..."

*sudo tee /etc/systemd/system/node_exporter.service > /dev/null
<<EOL*

[Unit]

Description=Node Exporter

After=network.target

[Service]

User=prometheus

ExecStart=/usr/local/bin/node_exporter

[Install]

WantedBy=multi-user.target

EOL

Step 5: Create a user for Node Exporter

echo "Creating a dedicated user for Node Exporter..."

sudo useradd --no-create-home --shell /bin/false prometheus

Step 6: Reload systemd, start, and enable Node Exporter

echo "Starting and enabling Node Exporter service..."

sudo systemctl daemon-reload

sudo systemctl start node_exporter

sudo systemctl enable node_exporter

echo "Node Exporter installation and configuration completed successfully!"

Now save and quit the file → Now give the file permission to execute the by using the command “***chmod +x install_node_exporter.sh***” → Now execute the file with the command “***./install_node_exporter.sh***” → Node Exporter metrics should now be available at “***http://<server_ip>:9100/metrics***” → Now we will setup Prometheus in grafana server for that connect the instance and create a .sh file using the command “***vi setup_prometheus_and_node_exporter.sh***” and type the following content inside it,

```
#!/bin/bash
```

```
# Exit script on any error
```

```
set -e
```

```
echo "Starting setup for Prometheus and Node Exporter..."
```

```
# --- Install Prometheus ---
```

```
echo "Downloading Prometheus..."
```

wget

*<https://github.com/prometheus/prometheus/releases/download/v2.47.0/prometheus-2.47.0.linux-amd64.tar.gz> -O
/tmp/prometheus.tar.gz*

echo "Extracting Prometheus..."

tar xvfz /tmp/prometheus.tar.gz -C /tmp/

echo "Moving Prometheus binaries to /usr/local/bin..."

*sudo mv /tmp/prometheus-2.47.0.linux-amd64/prometheus
/usr/local/bin/*

*sudo mv /tmp/prometheus-2.47.0.linux-amd64/promtool
/usr/local/bin/*

*echo "Moving Prometheus configuration and data
directories..."*

sudo mkdir -p /etc/prometheus /var/lib/prometheus

```
sudo mv /tmp/prometheus-2.47.0.linux-amd64/consoles  
/etc/prometheus/
```

```
sudo mv /tmp/prometheus-2.47.0.linux-  
amd64/console_libraries /etc/prometheus/
```

```
# --- Setup Prometheus configuration ---
```

```
echo "Creating Prometheus configuration file..."
```

```
sudo tee /etc/prometheus/prometheus.yml > /dev/null <<EOL
```

```
global:
```

```
  scrape_interval: 15s
```

```
scrape_configs:
```

```
  - job_name: 'node_exporter'
```

```
    static_configs:
```

```
      - targets: ['13.235.248.20:9100', '65.0.123.45:9100']
```

```
EOL
```


--- Create Prometheus systemd service ---

echo "Creating Prometheus systemd service..."

sudo tee /etc/systemd/system/prometheus.service > /dev/null

<<EOL

[Unit]

Description=Prometheus

Wants=network-online.target

After=network-online.target

[Service]

User=prometheus

ExecStart=/usr/local/bin/prometheus --

config.file=/etc/prometheus/prometheus.yml --

storage.tsdb.path=/var/lib/prometheus/ --

web.console.templates=/etc/prometheus/consoles --

web.console.libraries=/etc/prometheus/console_libraries

[Install]

WantedBy=multi-user.target

EOL

echo "Creating Prometheus user and setting permissions..."

sudo useradd --no-create-home --shell /bin/false prometheus

sudo chown -R prometheus:prometheus /etc/prometheus

/var/lib/prometheus

Start Prometheus

echo "Starting Prometheus service..."

sudo systemctl daemon-reload

sudo systemctl enable prometheus

sudo systemctl start prometheus

--- Install Node Exporter on Local Machine ---

echo "Installing Node Exporter on this machine..."

wget

*https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz -O
/tmp/node_exporter.tar.gz*

tar -xvzf /tmp/node_exporter.tar.gz -C /tmp/

*sudo mv /tmp/node_exporter-1.6.1.linux-amd64/node_exporter
/usr/local/bin/*

echo "Creating Node Exporter systemd service..."

*sudo tee /etc/systemd/system/node_exporter.service > /dev/null
<<EOL*

[Unit]

Description=Node Exporter

After=network.target

[Service]

User=prometheus

ExecStart=/usr/local/bin/node_exporter

[Install]

WantedBy=multi-user.target

EOL

echo "Starting Node Exporter service..."

sudo systemctl daemon-reload

sudo systemctl enable node_exporter

sudo systemctl start node_exporter

--- Completion ---

echo "Setup completed! Prometheus is running and Node Exporter is installed.""

Now give execute permissions to the file using command

“chmod +x setup_prometheus_and_node_exporter.sh” →

Execute the file using the command

“./setup_prometheus_and_node_exporter.sh” → Now the

Prometheus is accessible at <http://<grafana-server-ip>:9090> →

Go to Status → Target → To see the status of the targets → Now we will setup grafana for that create a .sh file using command “***vi install_grafana.sh***” and type the following content,

```
#!/bin/bash
```

```
# Update the package list
```

```
sudo apt-get update -y
```

```
# Install required software properties package
```

```
sudo apt-get install -y software-properties-common
```

```
# Add the Grafana repository
```

```
sudo add-apt-repository "deb [arch=amd64]
```

```
https://packages.grafana.com/oss/deb stable main"
```

```
# Add Grafana GPG key
```

```
wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -
```

Update the package list after adding the Grafana repository

sudo apt-get update -y

Install Grafana

sudo apt-get install -y grafana

Start Grafana service

sudo systemctl start grafana-server

Enable Grafana to start on boot

sudo systemctl enable grafana-server

Print success message

echo "Grafana has been installed and started successfully." →

Now the Grafana is accessible at “*http://<grafana-server-ip>:3000*”

Go to connections → Data sources → Prometheus → Specify Prometheus server URL as “*http://52.66.194.245:9090*” → Save and test →

Monitoring CPU Utilization

Go to home and click on “*Create your first dashboard*” → Click on “*Add visualization*” → Select metric as “*process_cpu_seconds_total*” → In the label filters choose “*instance*” and choose “*master-server IP*” → Select add query and choose metric as “*process_cpu_seconds_total*” → In the label filters choose “*instance*” and choose “*test-server IP*” → Choose the graph type that you want → Specify panel title as “*CPU Utilization*” → Save dashboard → Specify the Dashboard title as “*Monitoring*” → Save

Disk Space Utilization

Click on “*Add*” → Choose “*Visualization*” → Type metric as “*100 - (node_filesystem_avail_bytes{mountpoint="/" } * 100 / node_filesystem_size_bytes{mountpoint="/" })*” → It calculate the disk space utilization for both instances → Specify panel title as “*Disk Space Utilization*” → Add threshold value as



Green (0-80%): Normal usage.

Yellow (81-90%): Warning.

Red (91-100%): Critical.

Total Available Memory

Click on ***“Add”*** → Choose ***“Visualization”*** → Select metric as ***“node_memory_MemAvailable_bytes”*** → In the label filters choose ***“instance”*** and choose ***“master-server IP”*** → Select add query and choose metric as

“node_memory_MemAvailable_bytes” → In the label filters choose ***“instance”*** and choose ***“test-server IP”*** → Choose the graph type that you want → Specify panel title as ***“Total Available Memory”*** → Add threshold value as,

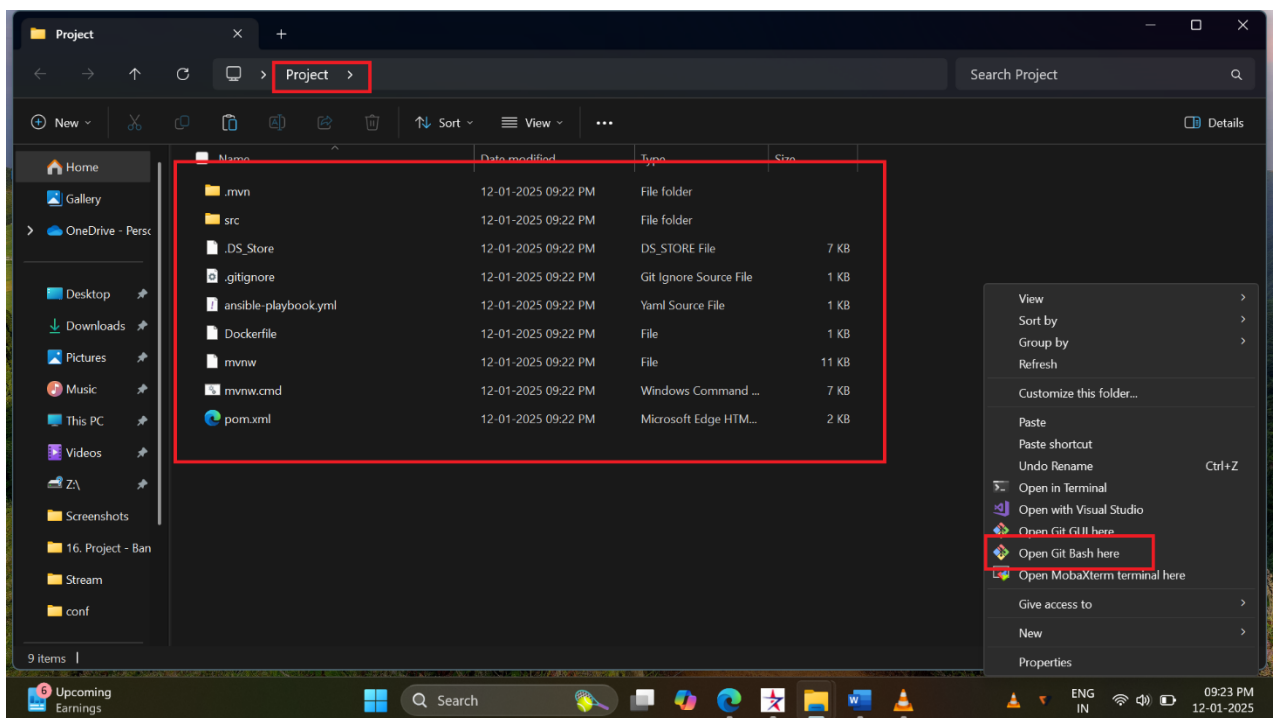
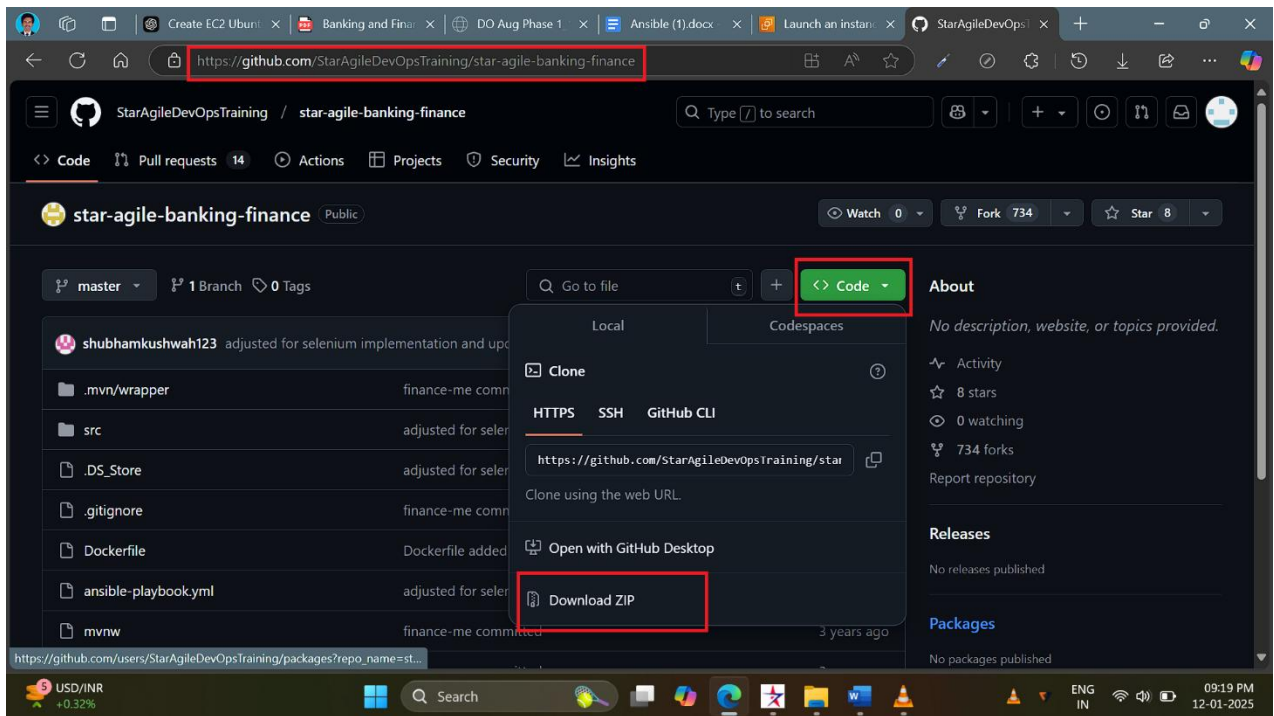
Green (0-80%): Normal usage.

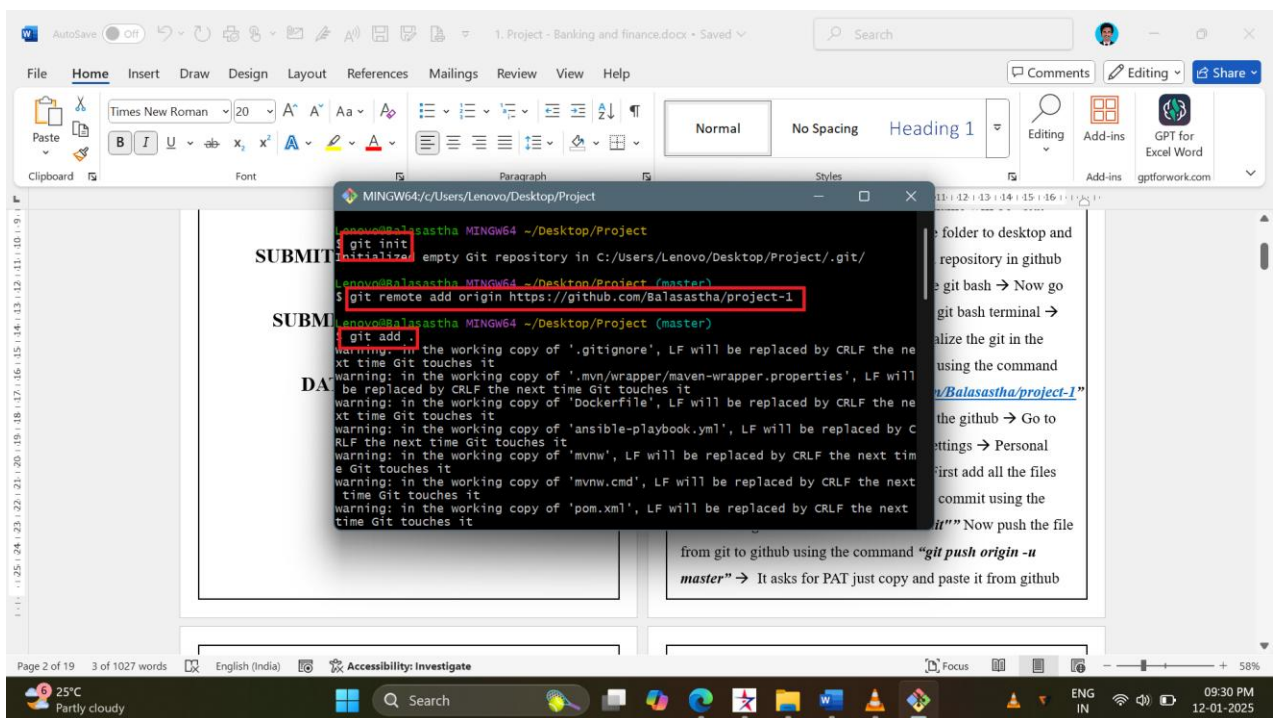
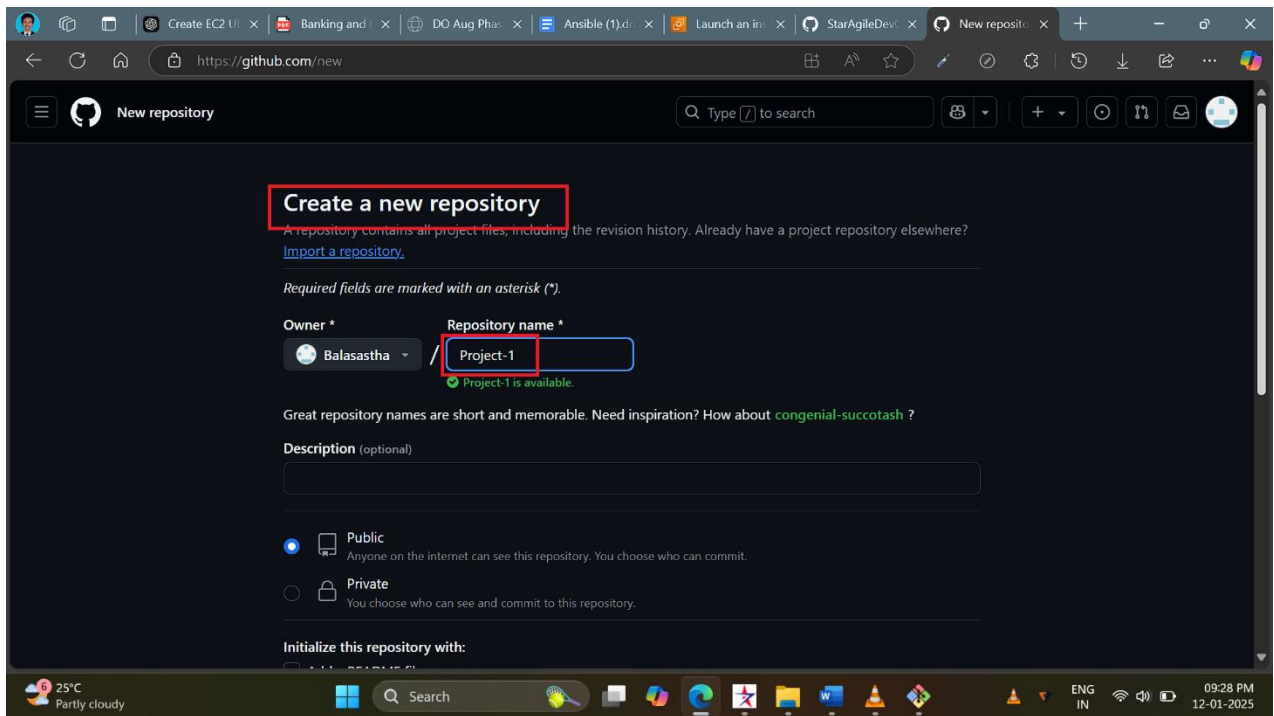
Yellow (81-90%): Warning.

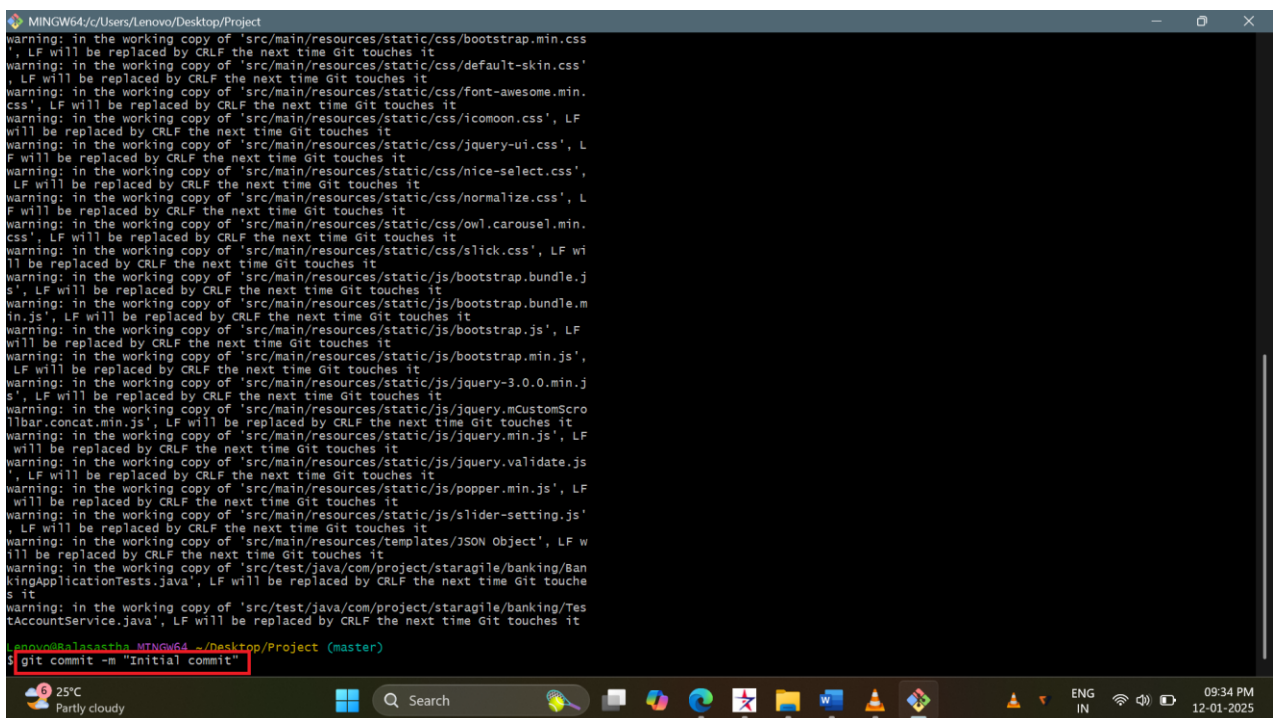
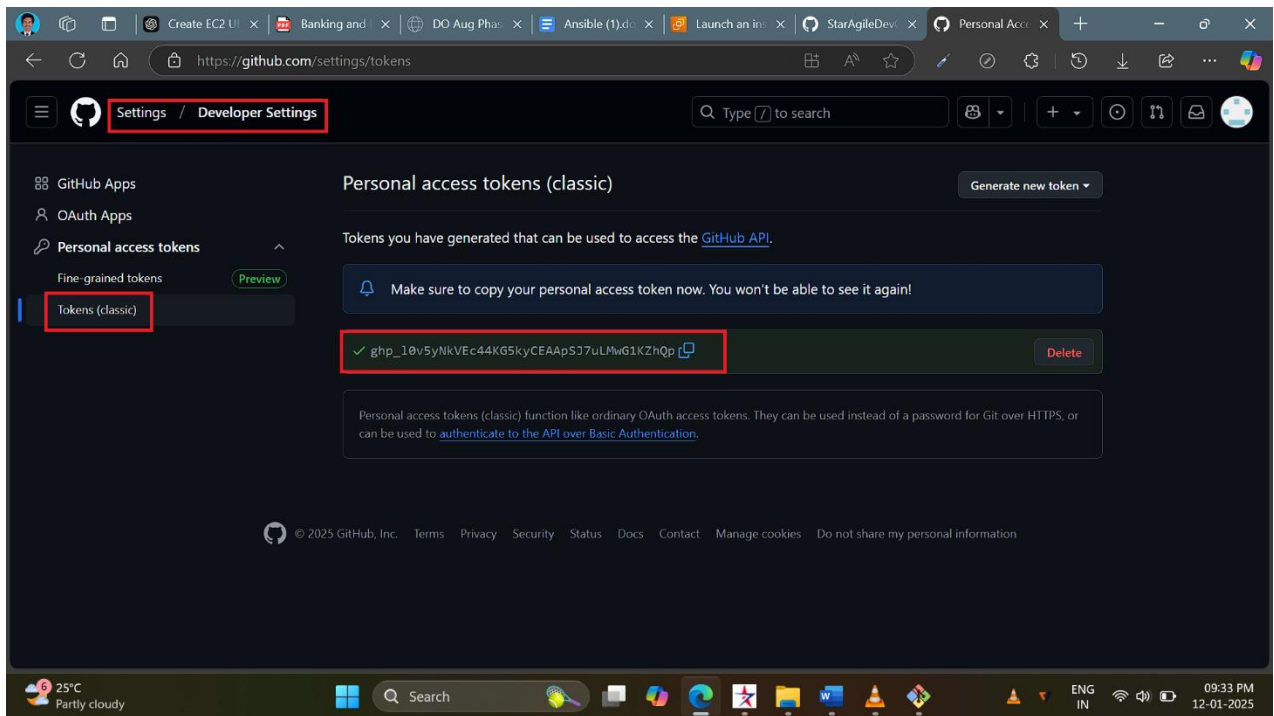
Red (91-100%): Critical.

Now whenever the developer push the code to github the jenkins job should be triggered automatically for that go to the github repository of the project → Go to settings → Webhooks

→ Add Webhooks → In the ***"Payload URL"*** field, provide the URL of your Jenkins server followed by /github-webhook/. For example, <http://your-jenkins-server/github-webhook/> → Set the content type to ***"application/json"*** → Under ***"Which events would you like to trigger this webhook?"***, select ***Just the push event*** → Save the webhook → Go to jenkins → Go to configure → Check ***"Github Project"*** and enter the github URL there and check ***"GitHub hook trigger for GITScm polling"*** → Now create a file in local repo and push it to the remote repo the job will be started automatically executed in the jenkins without any manual trigger.

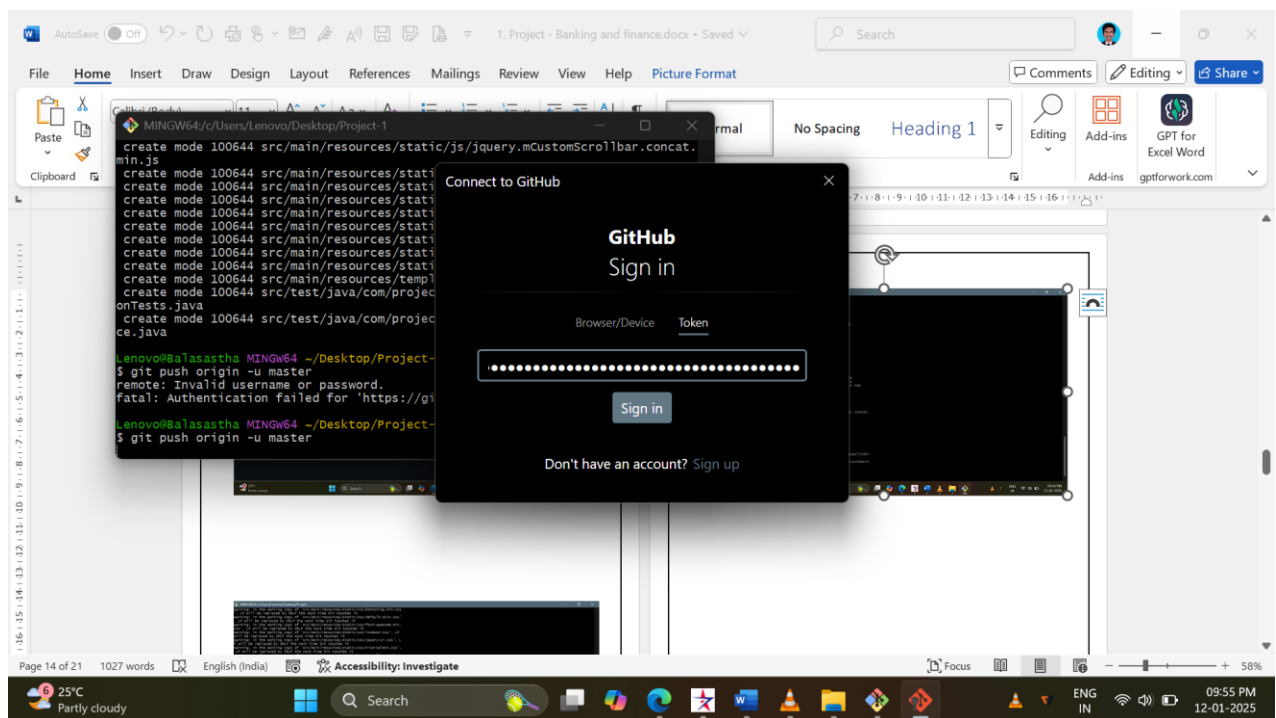


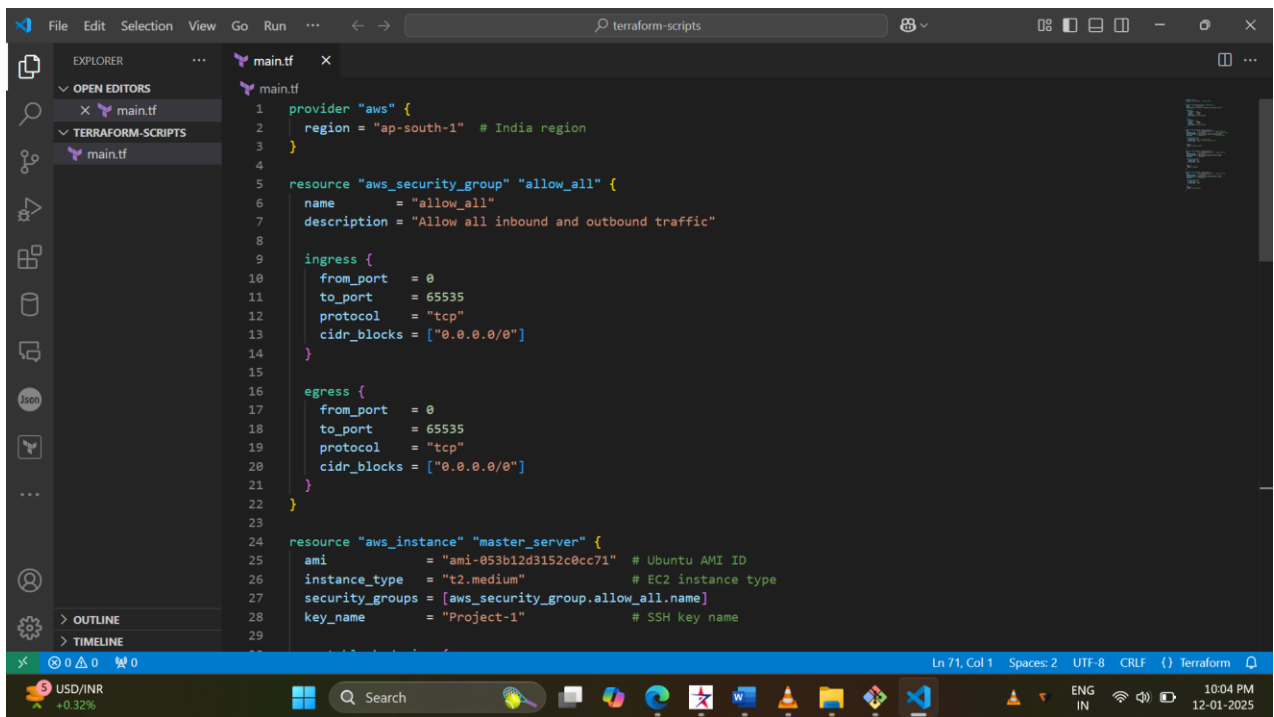
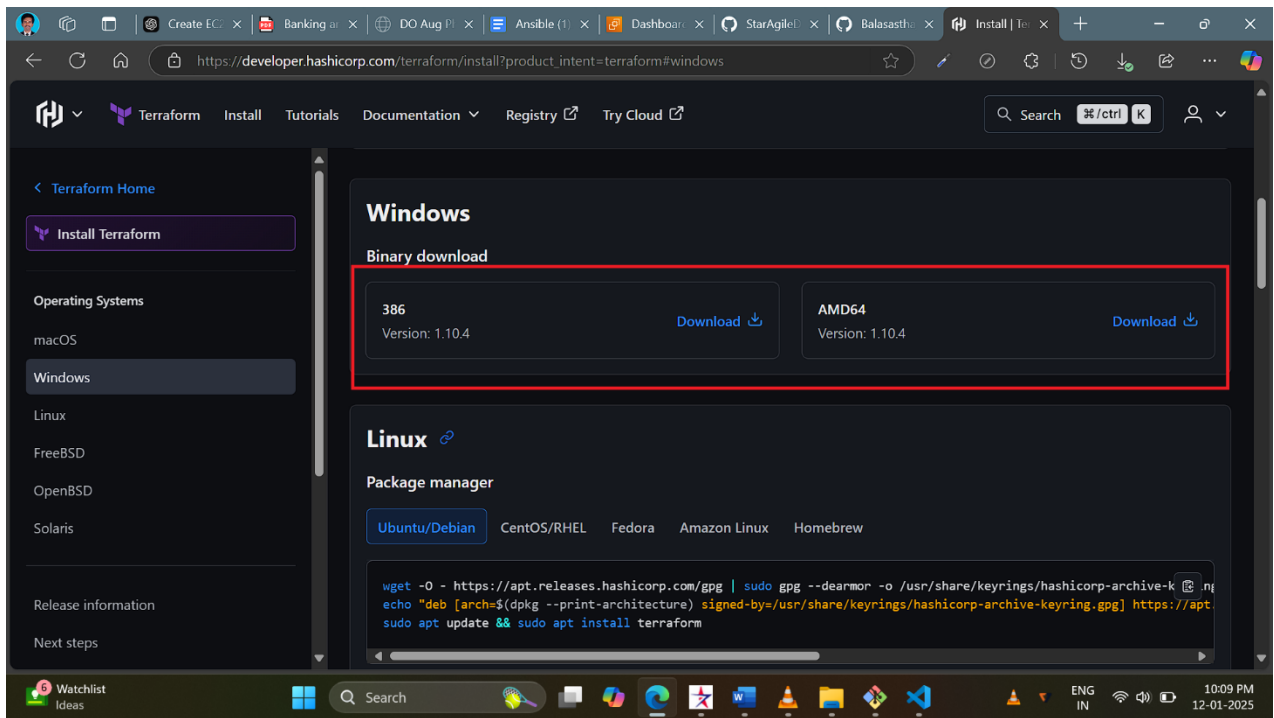


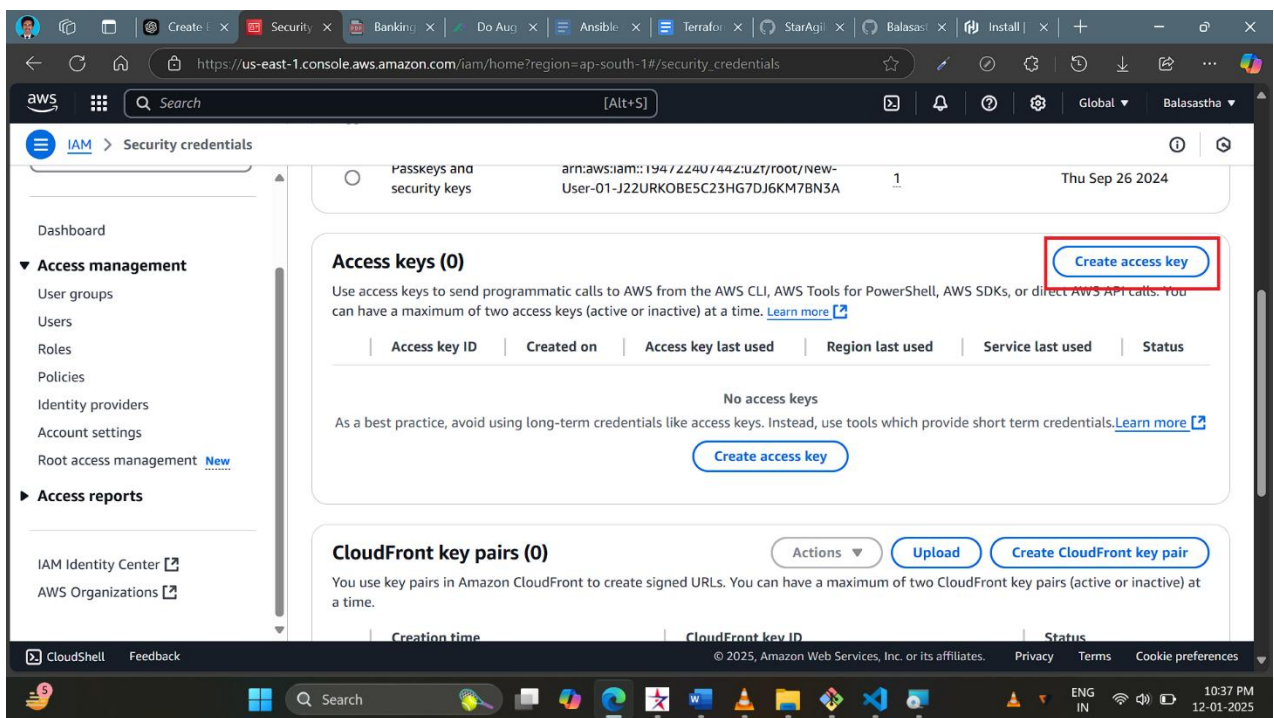
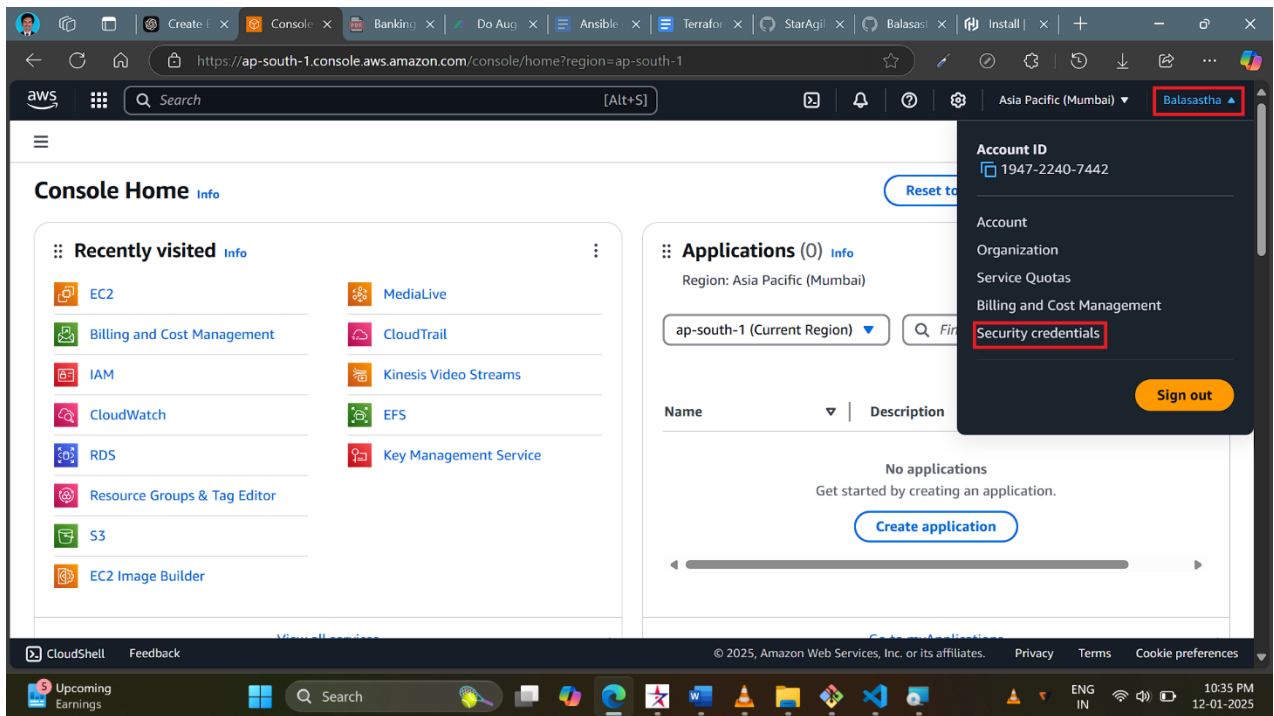


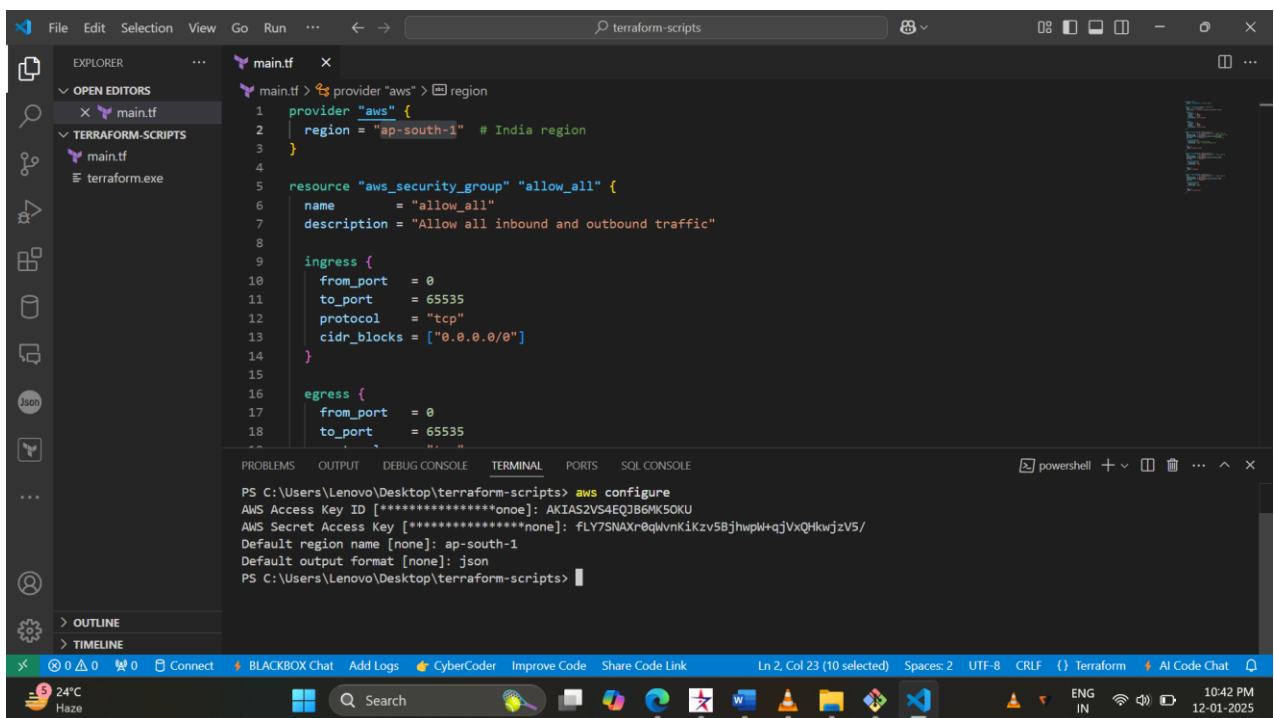
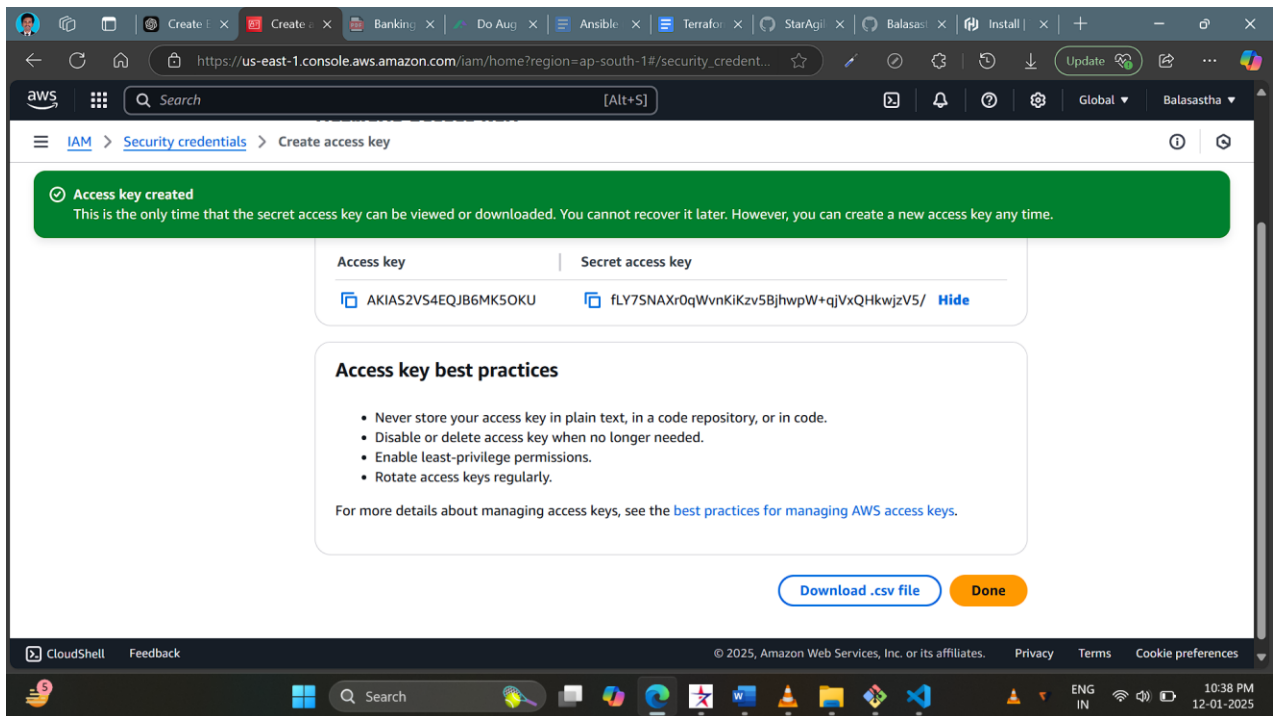
```
MINGW64/c:/Users/Lenovo/Desktop/Project-1
create mode 100644 src/main/resources/static/images/img-4.png
create mode 100644 src/main/resources/static/images/img-5.png
create mode 100644 src/main/resources/static/images/img-6.png
create mode 100644 src/main/resources/static/images/img-7.png
create mode 100644 src/main/resources/static/images/img-8.png
create mode 100644 src/main/resources/static/images/img-9.png
create mode 100644 src/main/resources/static/images/linkedin-icon.png
create mode 100644 src/main/resources/static/images/linkedin-icon1.png
create mode 100644 src/main/resources/static/images/logo.png
create mode 100644 src/main/resources/static/images/mail-icon.png
create mode 100644 src/main/resources/static/images/map-icon.png
create mode 100644 src/main/resources/static/images/round-1.png
create mode 100644 src/main/resources/static/images/round-2.png
create mode 100644 src/main/resources/static/images/search-icon.png
create mode 100644 src/main/resources/static/images/services-bg.png
create mode 100644 src/main/resources/static/images/twitter-icon.png
create mode 100644 src/main/resources/static/images/twitter-icon1.png
create mode 100644 src/main/resources/static/images/youtub-icon1.png
create mode 100644 src/main/resources/static/index.html
create mode 100644 src/main/resources/static/js/.DS_Store
create mode 100644 src/main/resources/static/js/bootstrap.bundle.js
create mode 100644 src/main/resources/static/js/bootstrap.bundle.js.map
create mode 100644 src/main/resources/static/js/bootstrap.bundle.min.js
create mode 100644 src/main/resources/static/js/bootstrap.bundle.min.js.map
create mode 100644 src/main/resources/static/js/bootstrap.js
create mode 100644 src/main/resources/static/js/bootstrap.js.map
create mode 100644 src/main/resources/static/js/bootstrap.min.js
create mode 100644 src/main/resources/static/js/bootstrap.min.js.map
create mode 100644 src/main/resources/static/js/custom.js
create mode 100644 src/main/resources/static/js/jquery-3.0.0.min.js
create mode 100644 src/main/resources/static/js/jquery.mCustomScrollbar.concat.min.js
create mode 100644 src/main/resources/static/js/jquery.min.js
create mode 100644 src/main/resources/static/js/jquery.validate.js
create mode 100644 src/main/resources/static/js/modernizer.js
create mode 100644 src/main/resources/static/js/plugin.js
create mode 100644 src/main/resources/static/js/popper.min.js
create mode 100644 src/main/resources/static/js/slider-setting.js
create mode 100644 src/main/resources/static/services.html
create mode 100644 src/main/resources/templates/team.html
create mode 100644 src/main/resources/templates/JSON Object
create mode 100644 src/test/java/com/project/staragile/banking/BankingApplicati
onTests.java
create mode 100644 src/test/java/com/project/staragile/banking/TestAccountServi
ce.java

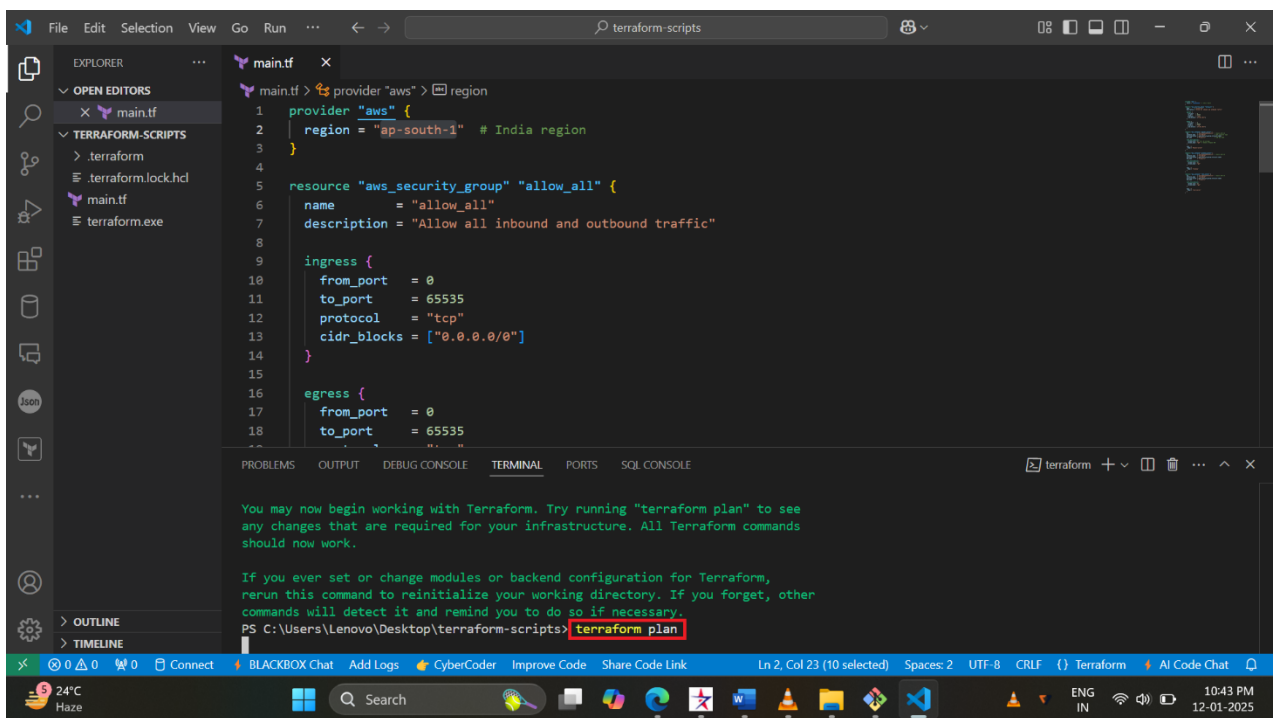
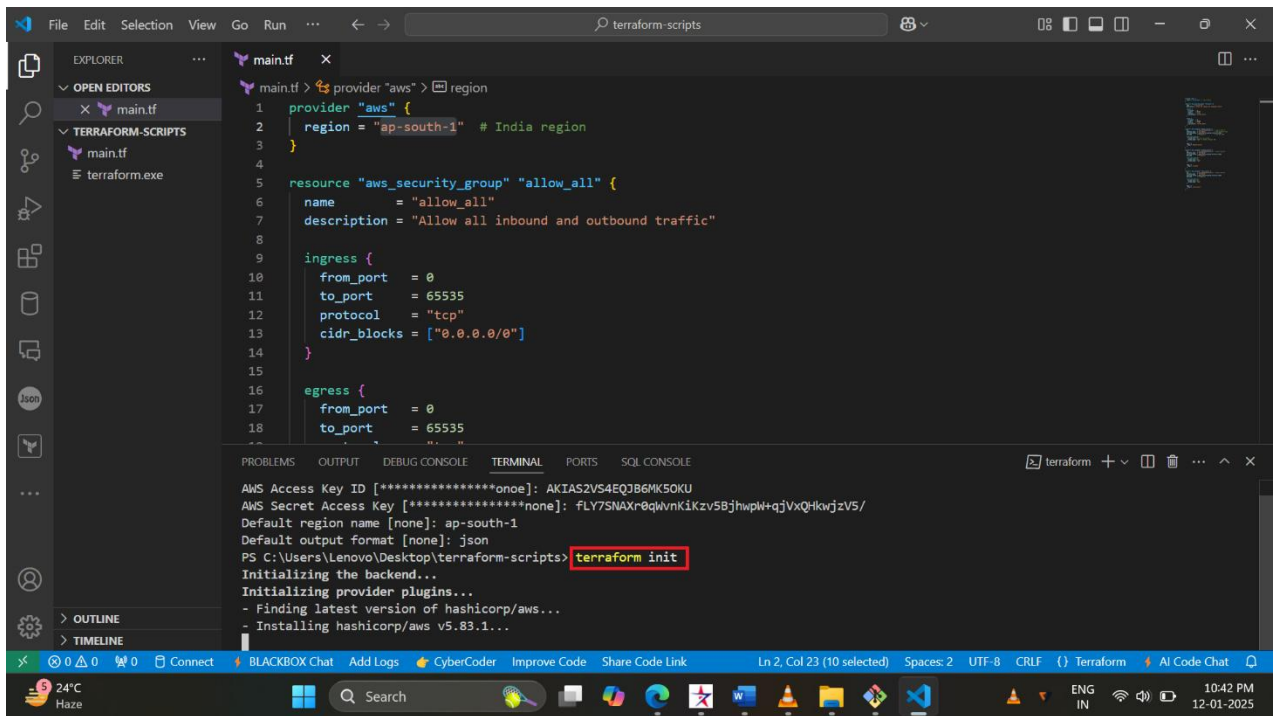
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git push origin -u master
```

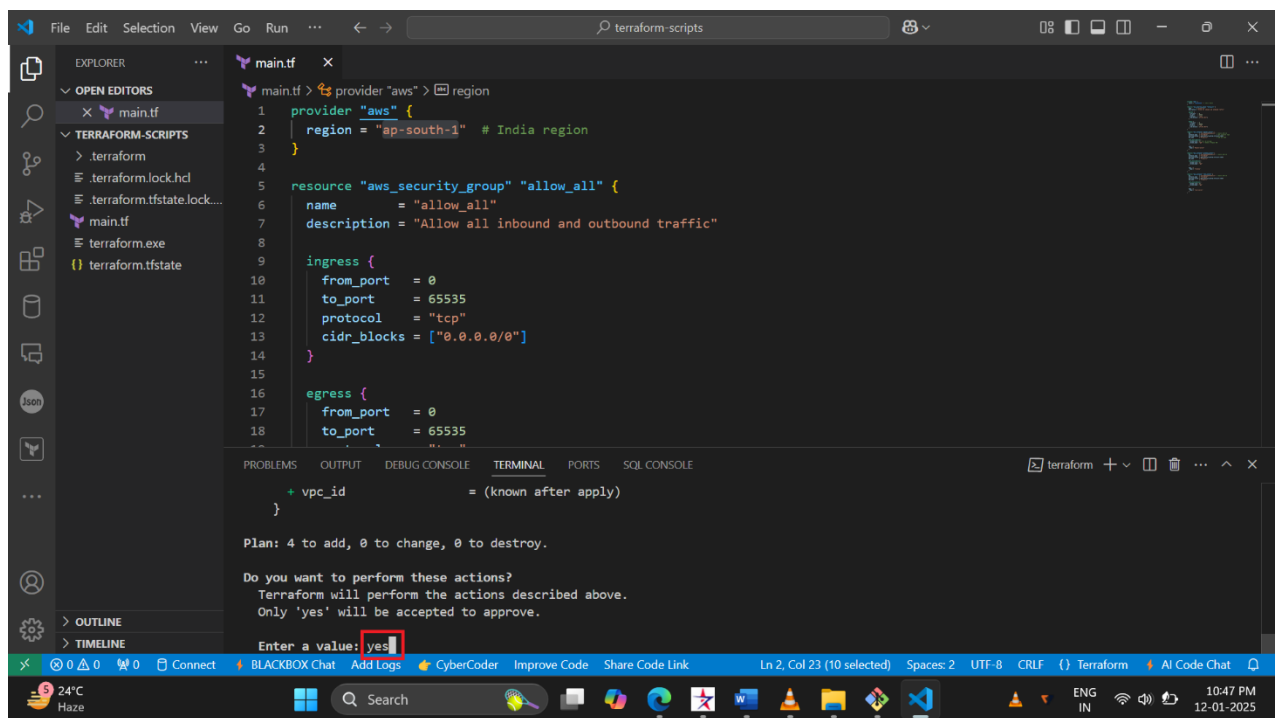
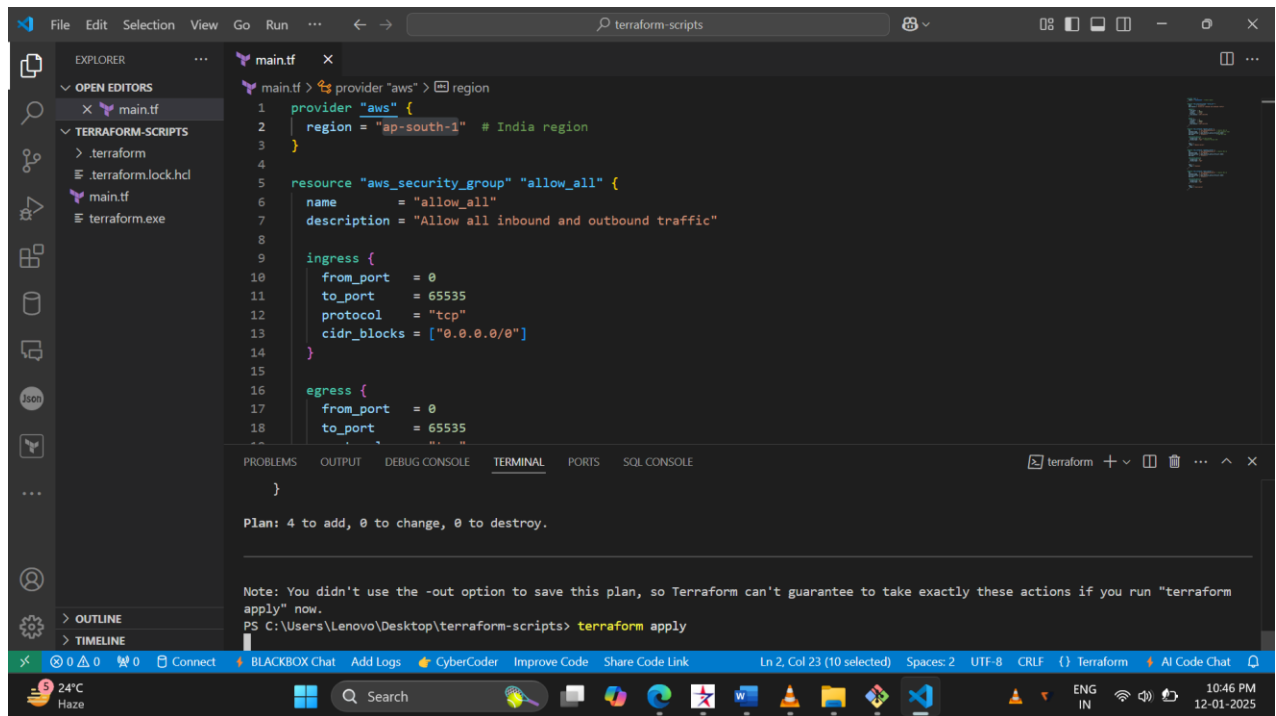












Instances (4) Info

Last updated 1 minute ago

Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Alarm
☐	Master-server	i-05ecea023a0b51fcf	Running	t2.medium	Initializing	View
☐	Grafana	i-0958f51b452a8a2fa	Running	t2.medium	Initializing	View
☐	Test-server	i-0414e6e27162befae	Running	t2.medium	Initializing	View

Select an instance

```
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jaotc to provide /usr/bin/jaotc (jaotc) in auto mode
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jhsdb to provide /usr/bin/jhsdb (jhsdb) in auto mode
Setting up openjdk-11-jdk:amd64 (11.0.25+9-lubuntu1~24.04) ...
update-alternatives: using /usr/lib/jvm/java-11-openjdk-amd64/bin/jconsole to provide /usr/bin/jconsole (jconsole) in auto mode
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

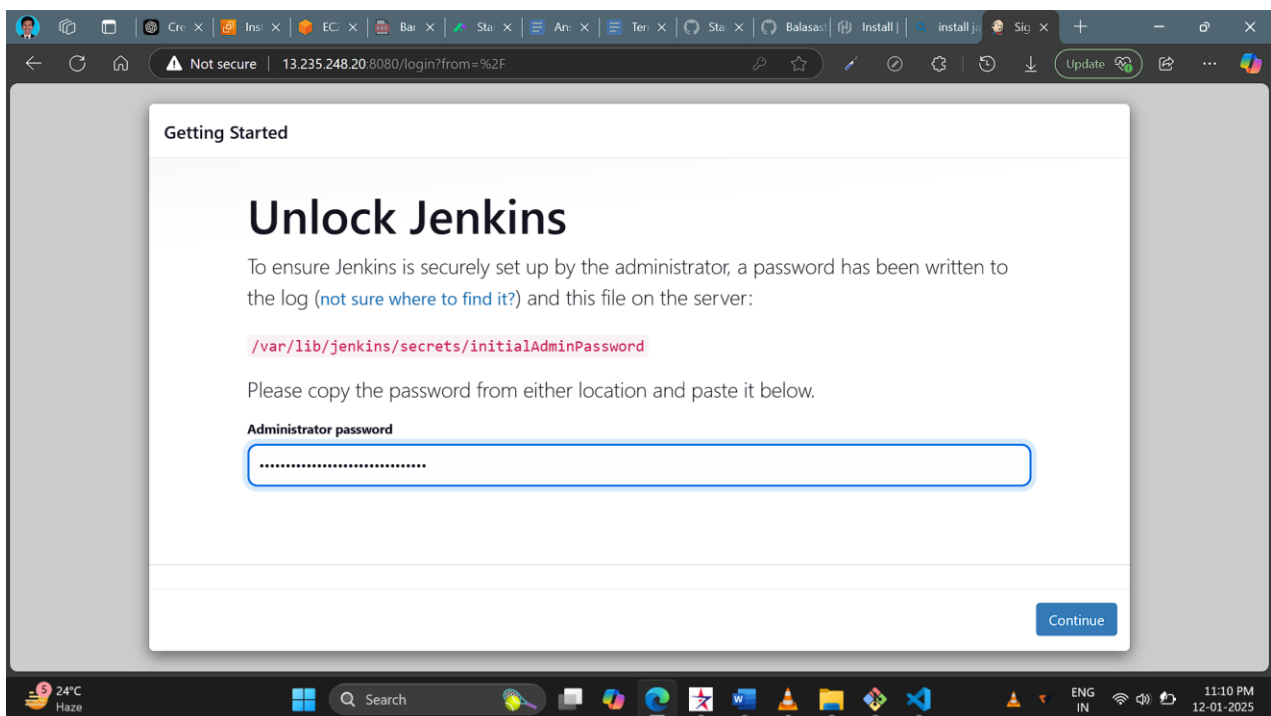
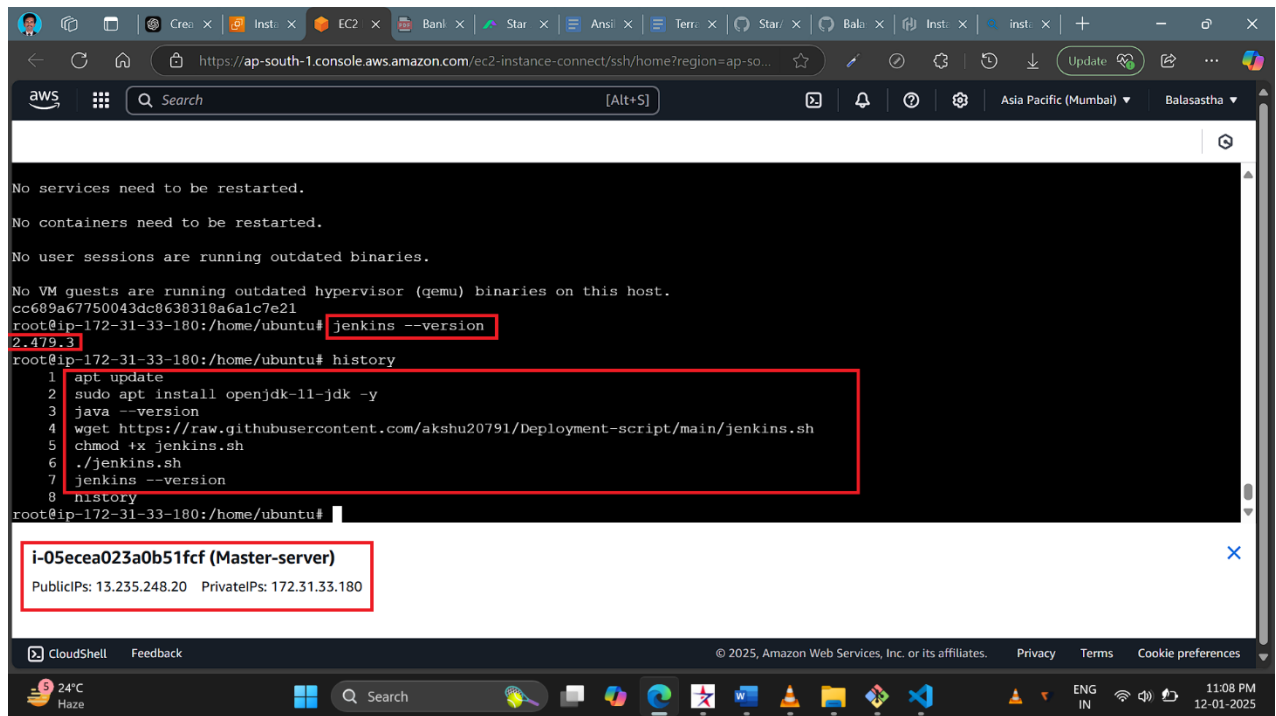
No containers need to be restarted.

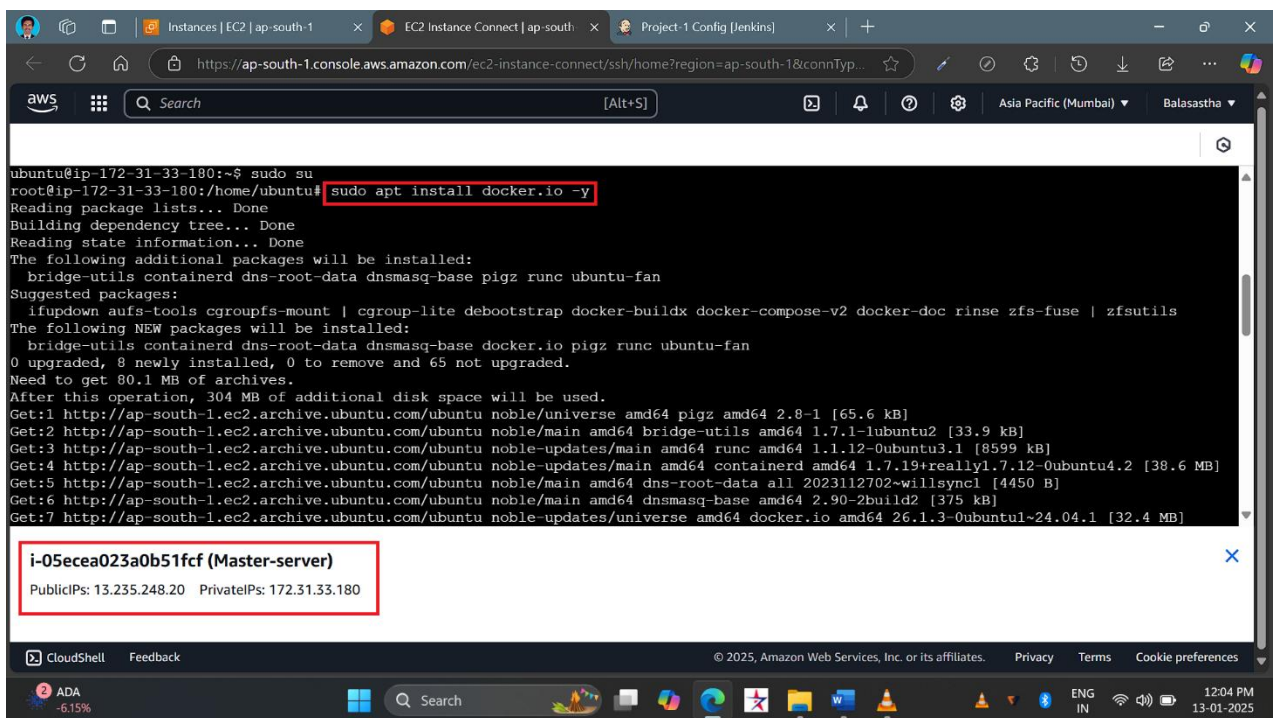
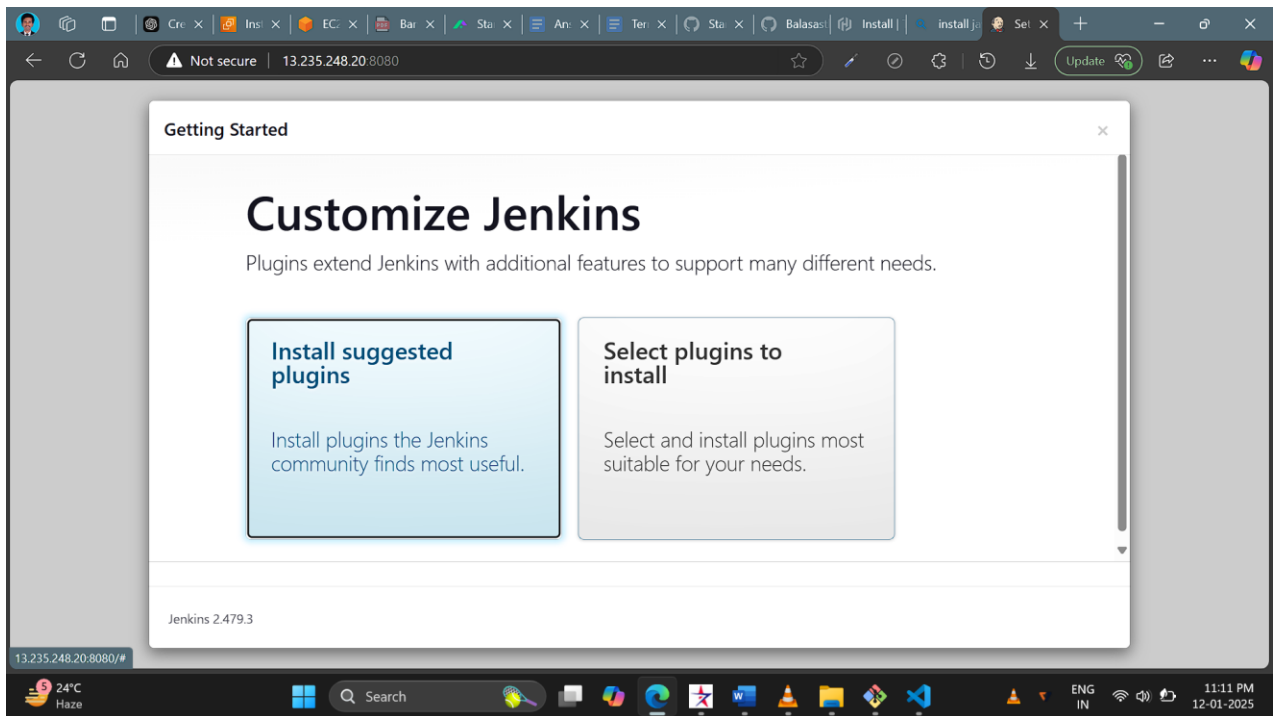
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-33-180:/home/ubuntu# java --version
openjdk 11.0.25 2024-10-15
OpenJDK Runtime Environment (build 11.0.25+9-post-Ubuntu-1ubuntu124.04)
OpenJDK 64-Bit Server VM (build 11.0.25+9-post-Ubuntu-1ubuntu124.04, mixed mode, sharing)
root@ip-172-31-33-180:/home/ubuntu#
```

i-05ecea023a0b51fcf (Master-server)

PublicIPs: 13.235.248.20 PrivateIPs: 172.31.33.180





Setting up docker.io (26.1.3-0ubuntu1~24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'docker' (GID 114) ...
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service → /usr/lib/systemd/system/docker.service.
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket → /usr/lib/systemd/system/docker.socket.
Processing triggers for dbus (1.14.10-4ubuntu4.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-33-180:/home/ubuntu# sudo usermod -aG docker jenkins
root@ip-172-31-33-180:/home/ubuntu# visudo

i-05ecea023a0b51fcf (Master-server)
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TATASTEEL -2.39%

Search

ENG IN 12:05 PM 13-01-2025

GNU nano 7.2 /etc/sudoers.tmp *

```
# User alias specification
# Cmnd alias specification
# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL
jenkins  ALL=(ALL:ALL) NOPASSWD: ALL
# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL

# See sudoers(5) for more information on "@include" directives:

@include /etc/sudoers.d
```

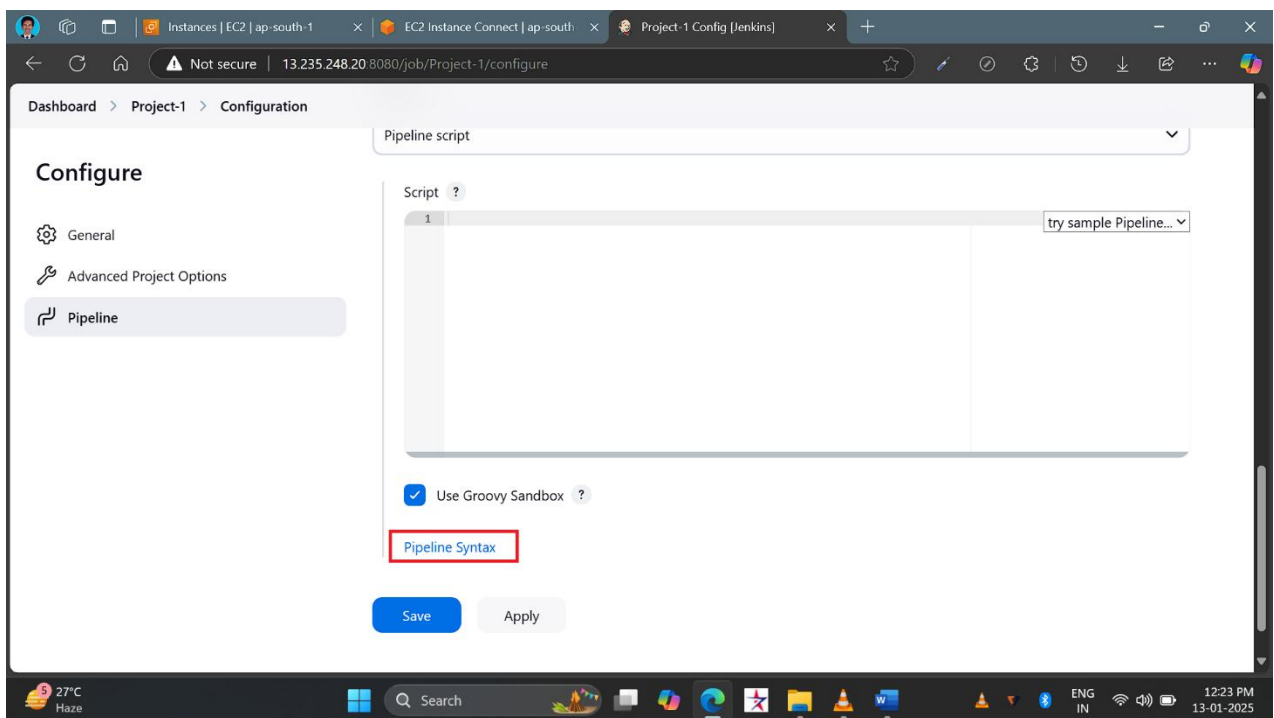
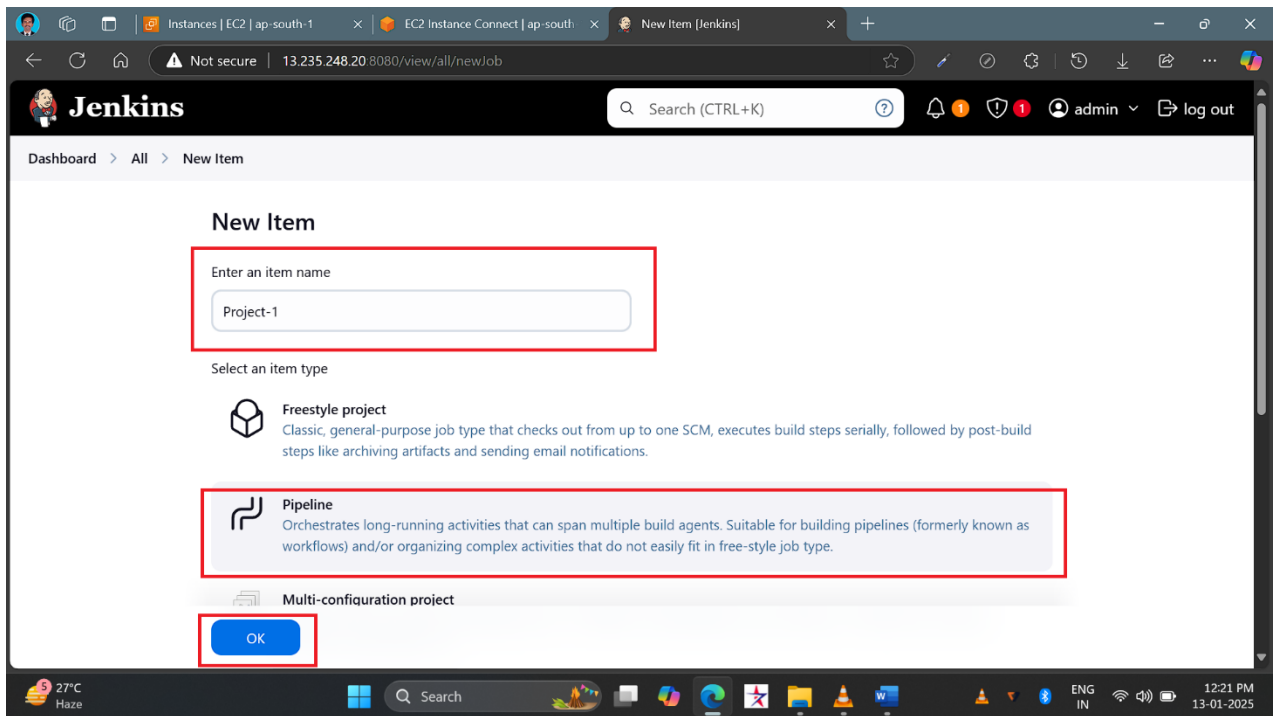
i-05ecea023a0b51fcf (Master-server)
PublicIPs: 13.235.248.20 PrivateIPs: 172.31.33.180

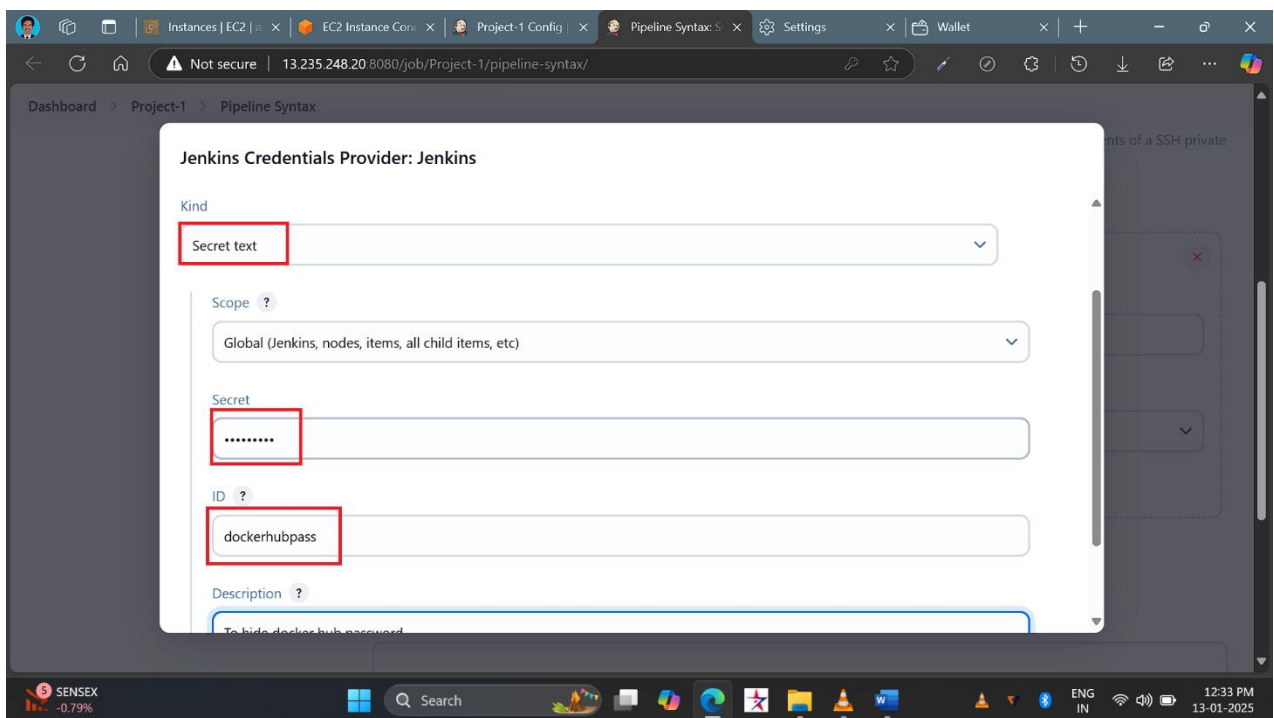
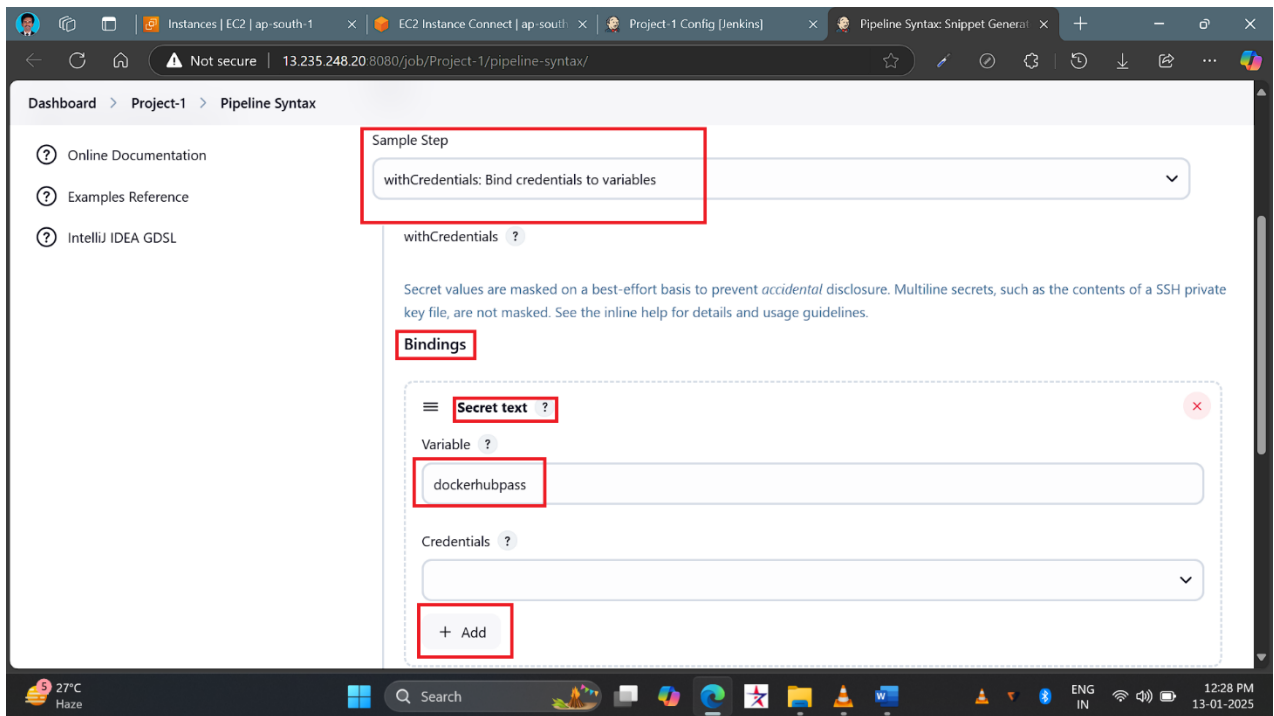
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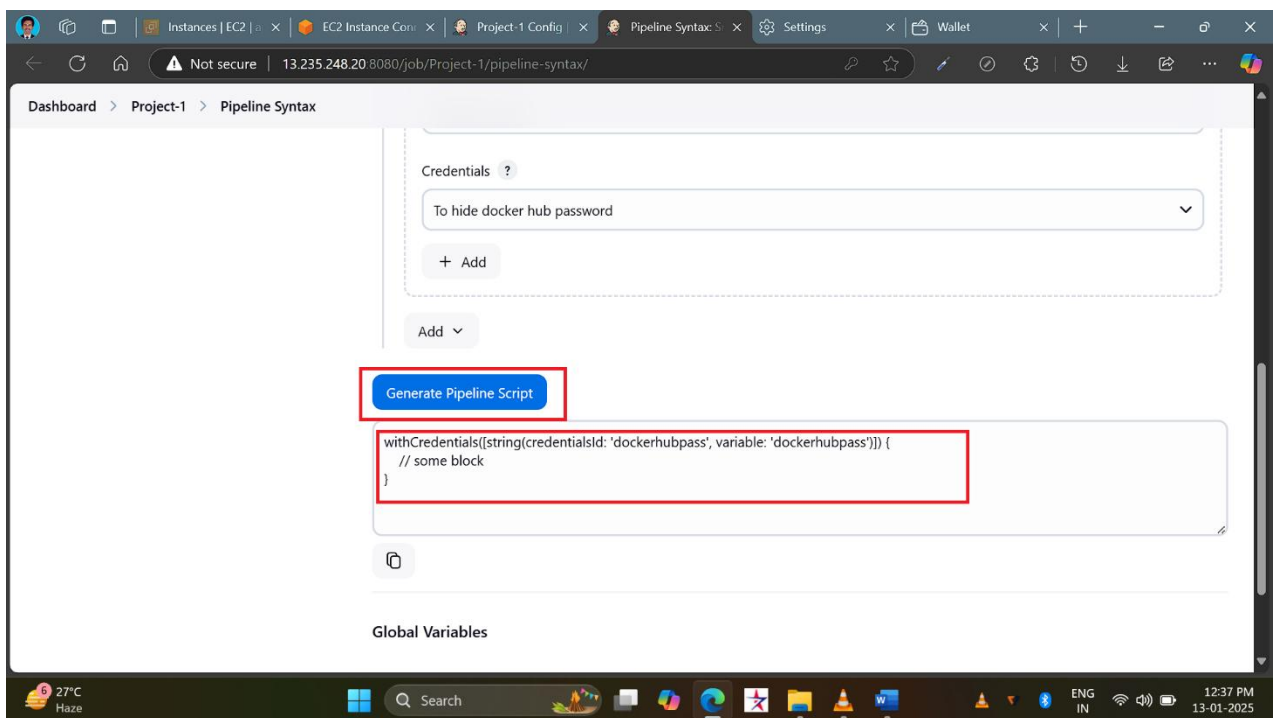
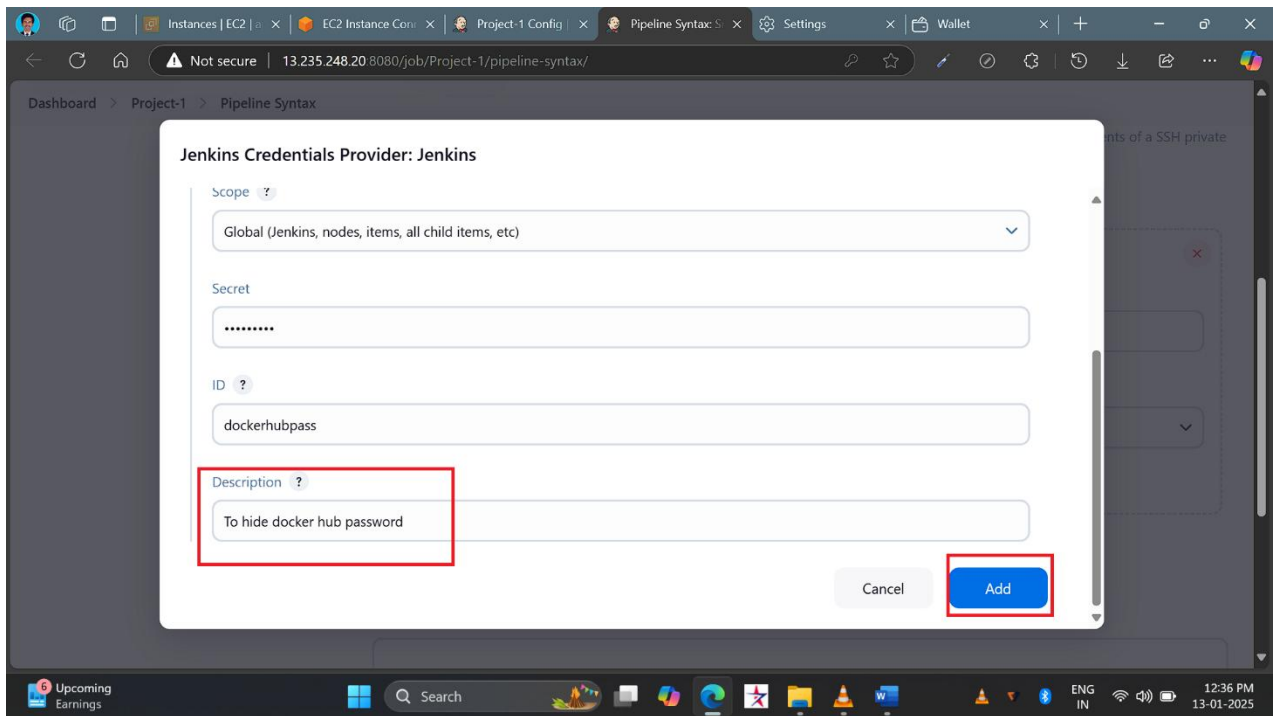
BSE midcap -2.96%

Search

ENG IN 12:07 PM 13-01-2025







root@ip-172-31-33-180:/home/ubuntu# **sudo apt install software-properties-common**

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  python3-software-properties
The following packages will be upgraded:
  python3-software-properties software-properties-common
2 upgraded, 0 newly installed, 0 to remove and 63 not upgraded.
Need to get 44.1 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common all 0.99.49.1 [14.4 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties all 0.99.49.1 [29.7 kB]
Fetched 44.1 kB in 0s (2326 kB/s)
(Reading database ... 87995 files and directories currently installed.)
Preparing to unpack .../software-properties-common 0.99.49.1 all.deb ...
Unpacking software-properties-common (0.99.49.1) over (0.99.48) ...
Preparing to unpack .../python3-software-properties 0.99.49.1 all.deb ...
Unpacking python3-software-properties (0.99.49.1) over (0.99.48) ...
Setting up python3-software-properties (0.99.49.1) ...
```

i-05ecea023a0b51fcf (Master-server)
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No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host

root@ip-172-31-33-180:/home/ubuntu# **sudo add-apt-repository --yes --update ppa:ansible/ansible**

```
Repository: 'Types: deb
URIs: https://ppa.launchpadcontent.net/ansible/ansible/ubuntu/
Suites: noble
Components: main
'
Description:
Ansible is a radically simple IT automation platform that makes your applications and systems easier to deploy. Avoid writing scripts or custom code to deploy and update your applications- automate in a language that approaches plain English, using SSH, with no agents to install on remote systems.

http://ansible.com/

If you face any issues while installing Ansible PPA, file an issue here:
https://github.com/ansible-community/ppa/issues
More info: https://launchpad.net/~ansible/+archive/ubuntu/ansible
Adding repository.
```

i-05ecea023a0b51fcf (Master-server)
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NIFTY -0.89%

Search

01:07 PM 13-01-2025

aws console terminal output:

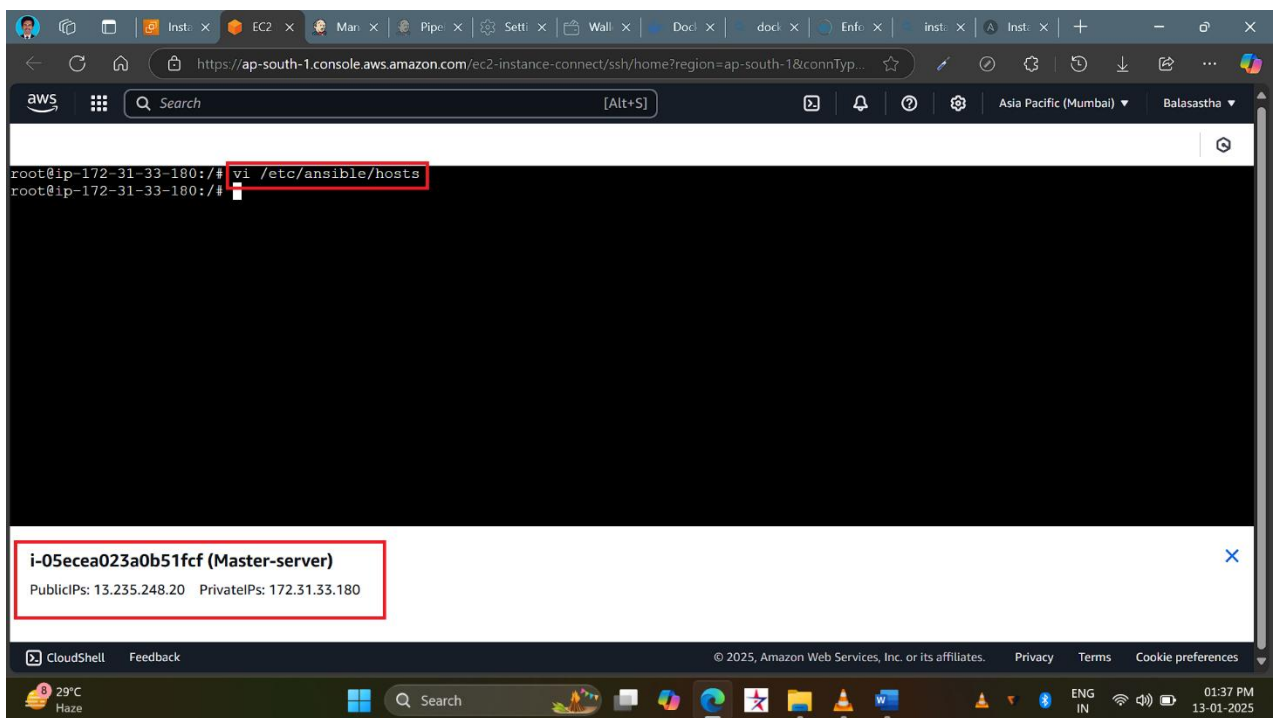
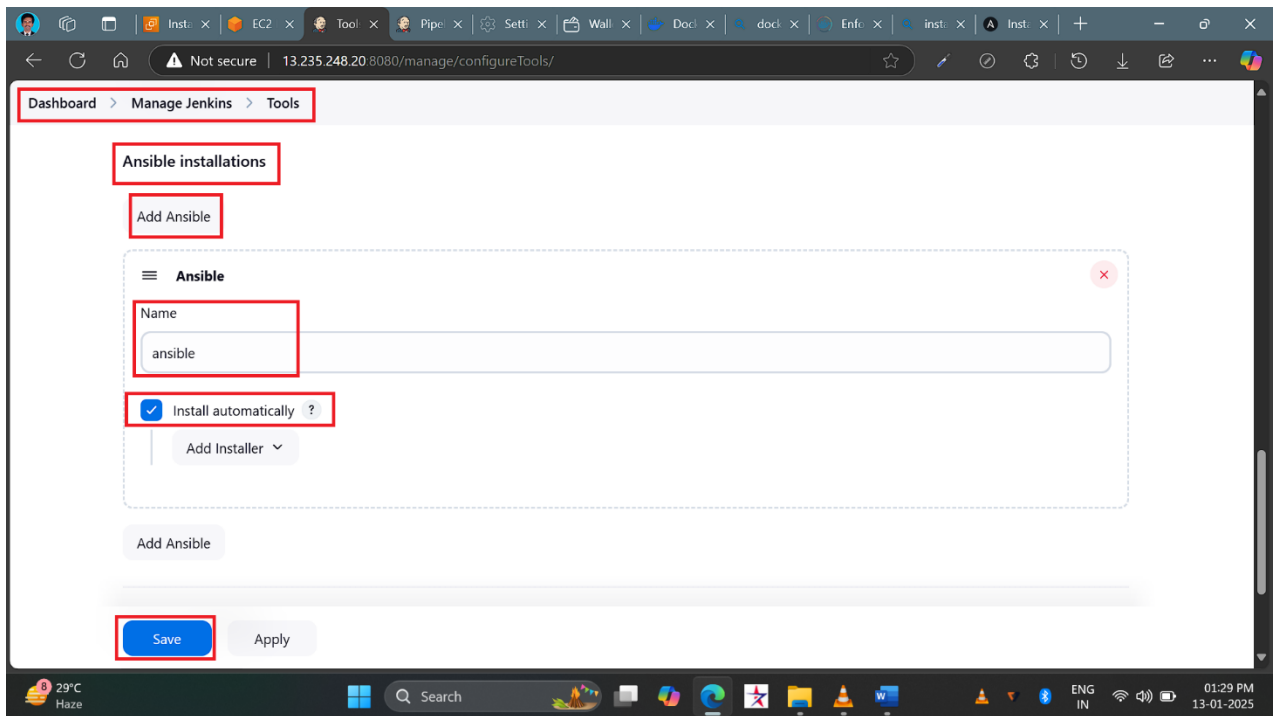
```
Fetched 145 kB in 1s (113 kB/s)
Reading package lists... Done
root@ip-172-31-33-180:/home/ubuntu# sudo apt install ansible
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  ansible-core python3-kerberos python3-nacl python3-ntlm-auth python3-paramiko python3-requests-ntlm python3-resolvelib python3-winrm
  python3-xmltodict sshpass
Suggested packages:
  python-nacl-doc python3-gssapi python3-invoke
The following NEW packages will be installed:
  ansible ansible-core python3-kerberos python3-nacl python3-ntlm-auth python3-paramiko python3-requests-ntlm python3-resolvelib
  python3-winrm python3-xmltodict sshpass
0 upgraded, 11 newly installed, 0 to remove and 63 not upgraded.
Need to get 19.2 MB of archives.
After this operation, 213 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-resolvelib all 1.0.1-1 [25.7 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 python3-kerberos amd64 1.1.14-3.1build9 [21.2 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 python3-nacl amd64 1.5.0-4build1 [57.9 kB]
```

i-05ecea023a0b51fcf (Master-server)
PublicIPs: 13.235.248.20 PrivateIPs: 172.31.33.180

Jenkins interface:

- Navigation: **Dashboard > Manage Jenkins > Plugins**
- Left sidebar: **Available plugins** (selected)
- Search: Search available plugins
- Buttons: **Install** (highlighted)
- Table of available plugins:

Install	Name	Released
☒	Docker 1.7.0 Cloud Providers Cluster Management docker This plugin integrates Jenkins with Docker	3 mo 0 days ago
☒	Ansible 403.v8d0ca_dcb_b_502 pipeline External Site/Tool Integrations DevOps Build Tools Deployment Invoke Ansible Ad-Hoc commands and playbooks. <i>This plugin is up for adoption! We are looking for new maintainers. Visit our Adopt a Plugin initiative for more information.</i>	6 mo 29 days ago
☐	JavaMail API 1.6.2-10 Library plugins (for use by other plugins)	7 mo 24 days ago



aws console window showing Ansible host configuration examples and a terminal window showing the command to edit the hosts file.

```
# - Comments begin with the '#' character
# - Blank lines are ignored
# - Groups of hosts are delimited by [header] elements
# - You can enter hostnames or ip addresses
# - A hostname/ip can be a member of multiple groups

# Ex 1: Ungrouped hosts, specify before any group headers:

## green.example.com
## blue.example.com
## 192.168.100.1
## 192.168.100.10
[ansiblegroup]
172.31.36.228

# Ex 2: A collection of hosts belonging to the 'webservers' group:

## [webservers]
## alpha.example.org
## beta.example.org
"/etc/ansible/hosts" 55L, 1203B
```

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29°C Haze 01:38 PM 13-01-2025

Terminal window showing the command to edit the hosts file.

```
MINGW64/c/Users/Lenovo/Desktop/Project-1
C:\Users\Lenovo\Desktop\Project-1 (master)
$ vi ansible-playbook.yml
```

Watchlist -2.57% 02:21 PM 13-01-2025

```
MINGW64/c/Users/Lenovo/Desktop/Project-1
- name: Configure Docker on EC2 Instances
  hosts: all
  become: true
  connection: ssh
  tasks:
    - name: updating apt
      command: sudo apt-get update

    - name: Install Docker
      command: sudo apt-get install -y docker.io

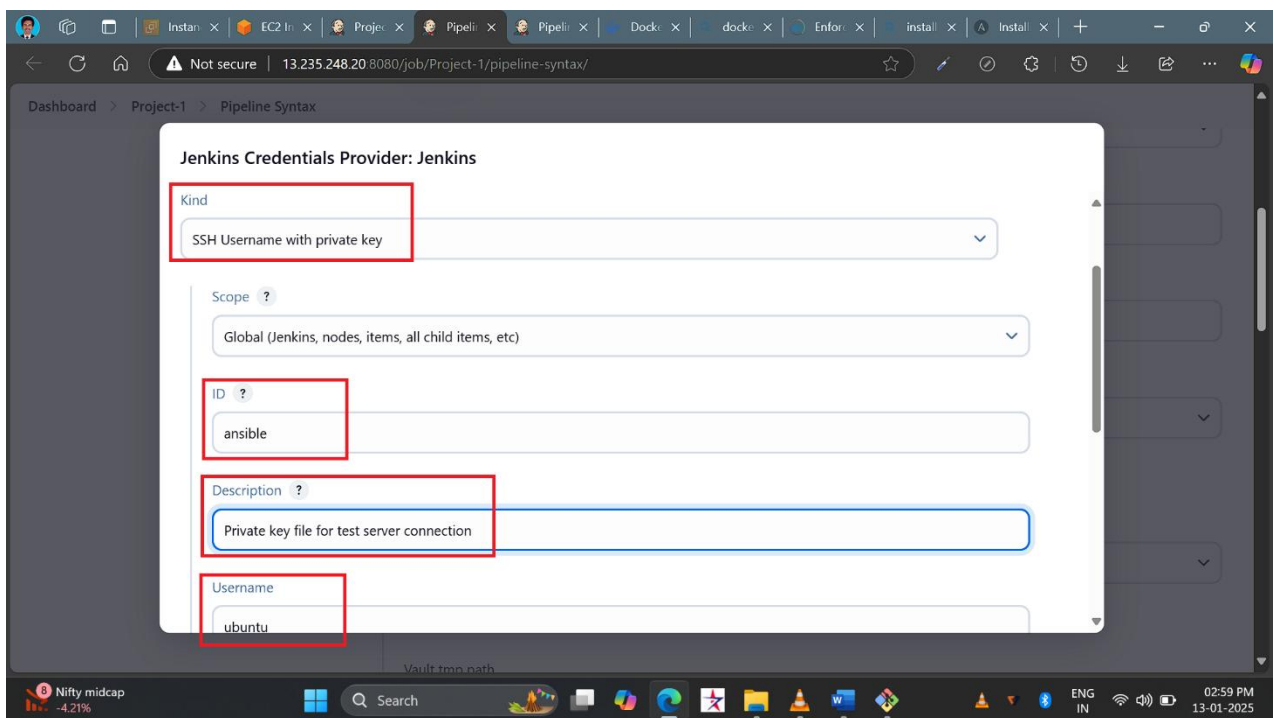
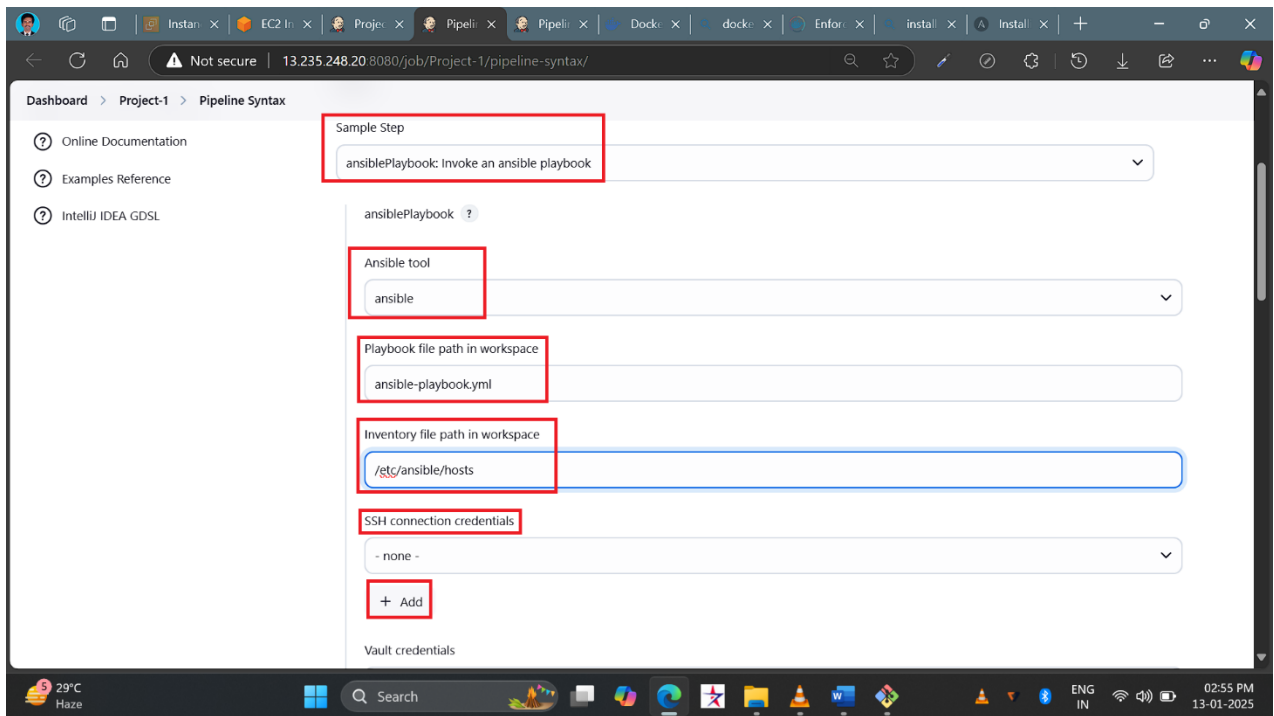
    - name: Start Docker Service
      command: sudo systemctl start docker

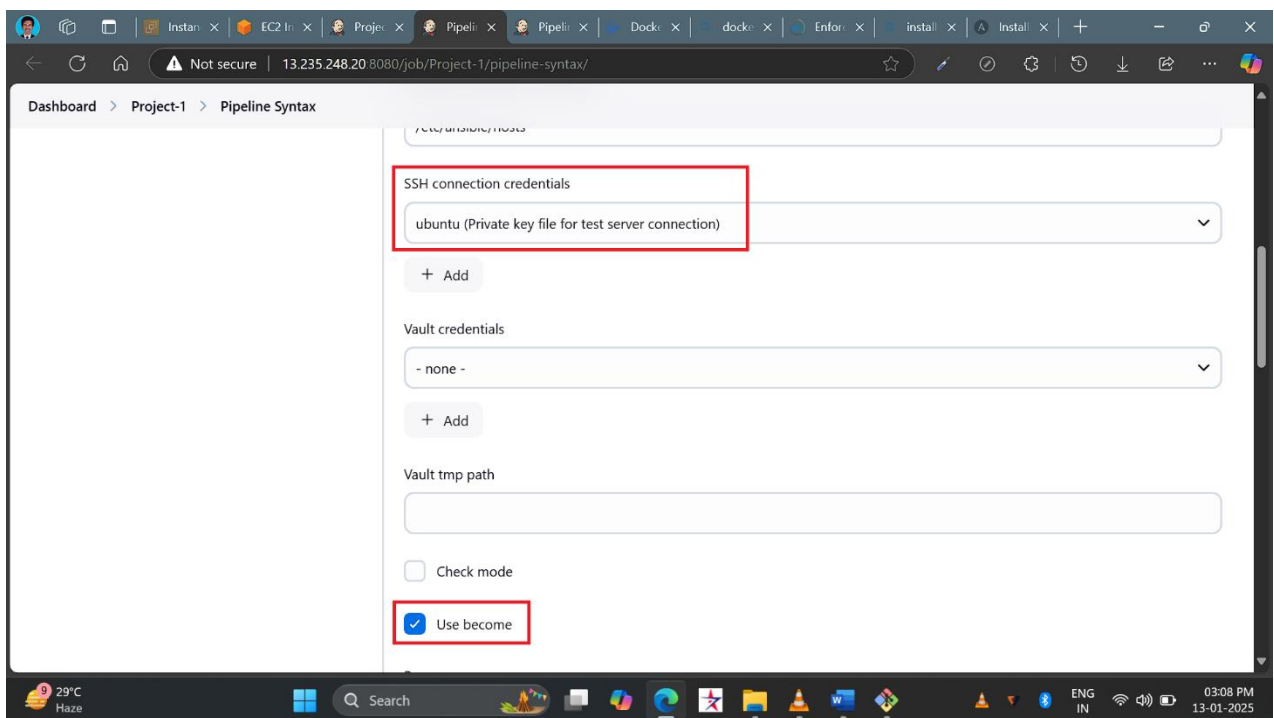
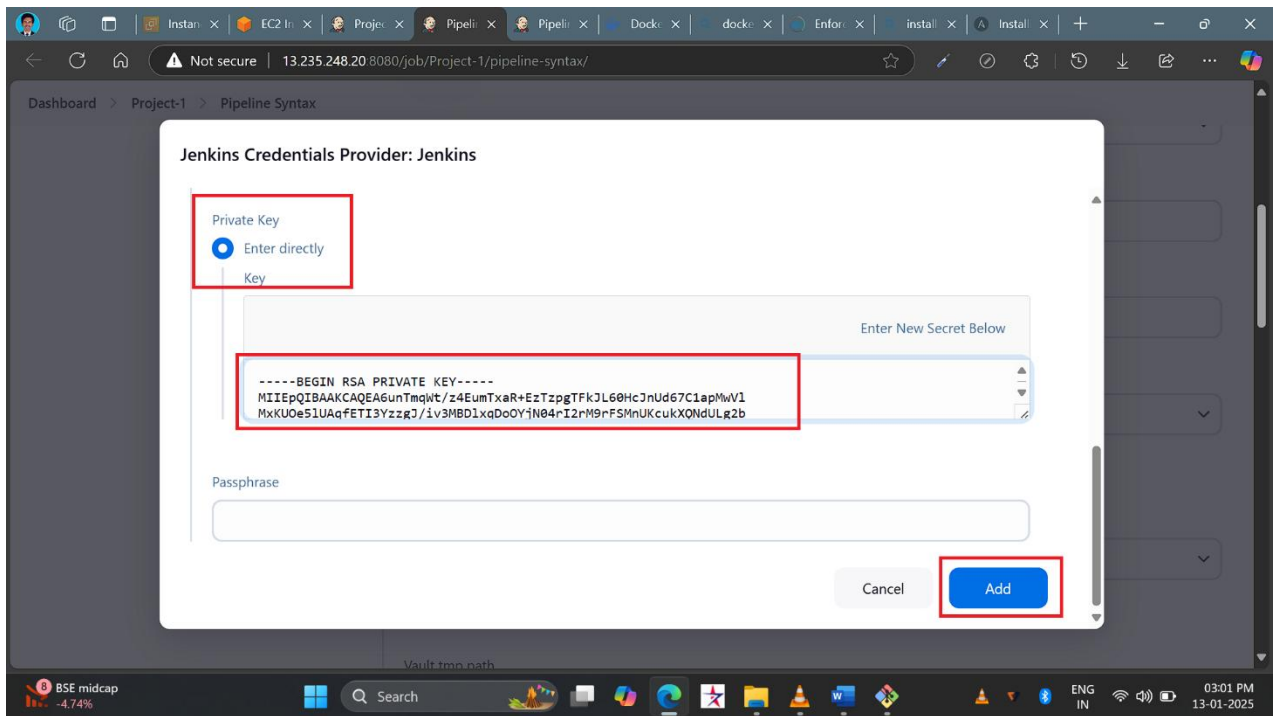
    - name: Deploy Docker Container
      command: docker run -itd -p 8084:8081 balasastha/banking-finance:1

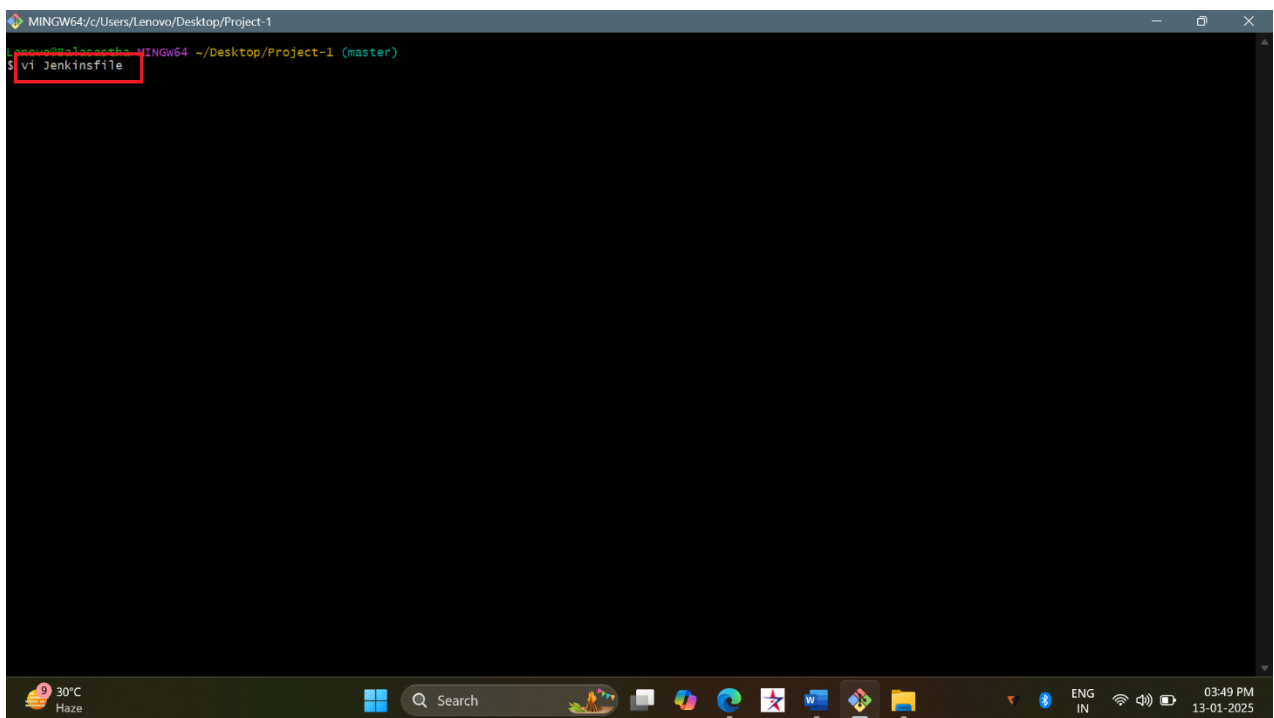
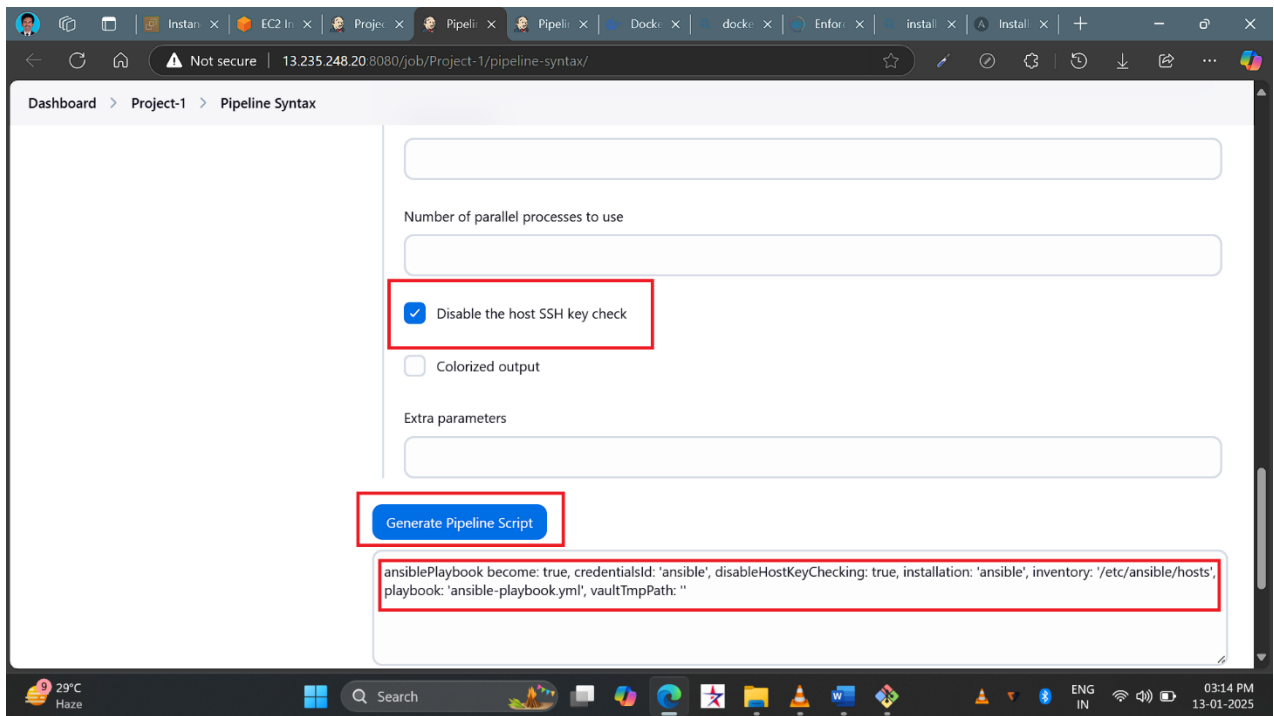
ansible-playbook.yml [unix] (14:18 13/01/2025) 16,70 All
"ansible-playbook.yml" [unix] 16L, 415B

6 Watchlist -2.57%
```

```
MINGW64/c/Users/Lenovo/Desktop/Project-1
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ vi ansible-playbook.yml
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git add ansible-playbook.yml
warning: in the working copy of 'ansible-playbook.yml', LF will be replaced by CRLF the next time Git touches it
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git commit -m "Second commit"
[master 35c8625] Second commit
1 file changed, 1 insertion(+), 1 deletion(-)
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git push origin -u master
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 312 bytes | 156.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Balasastha/Project-1
    1ccd0b2..35c8625 master -> master
branch 'master' set up to track 'origin/master'.
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$
```







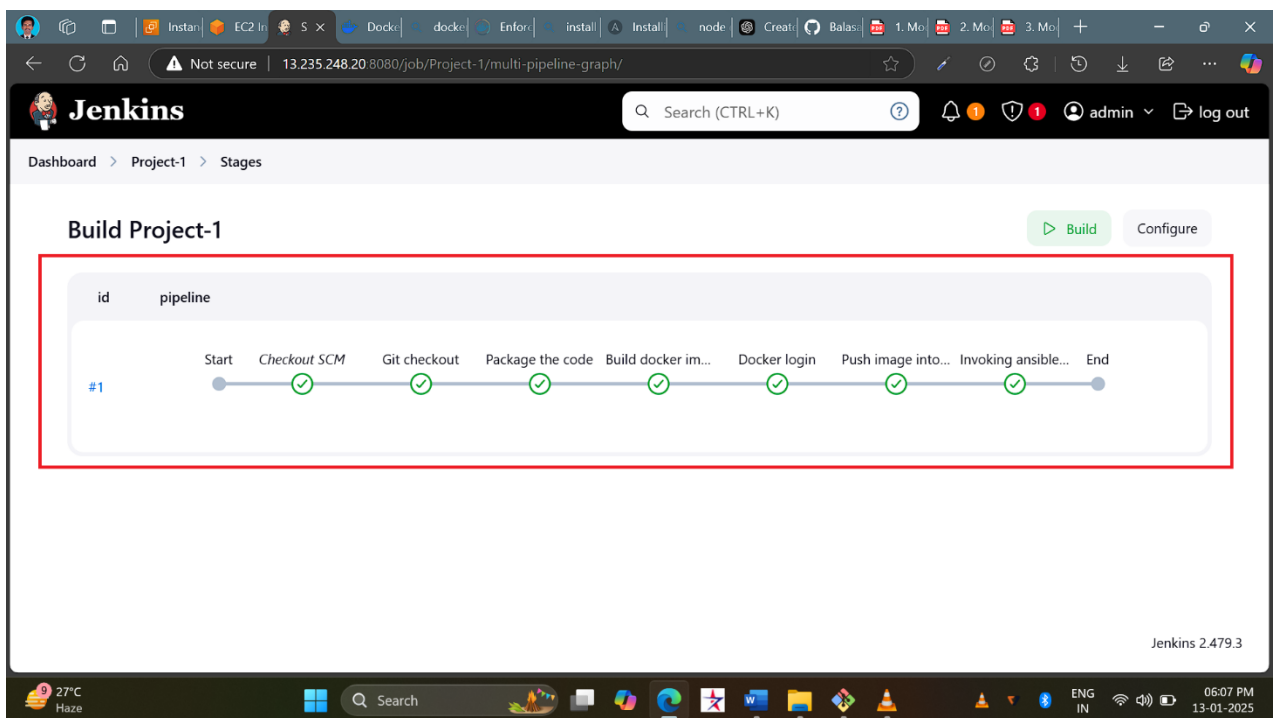
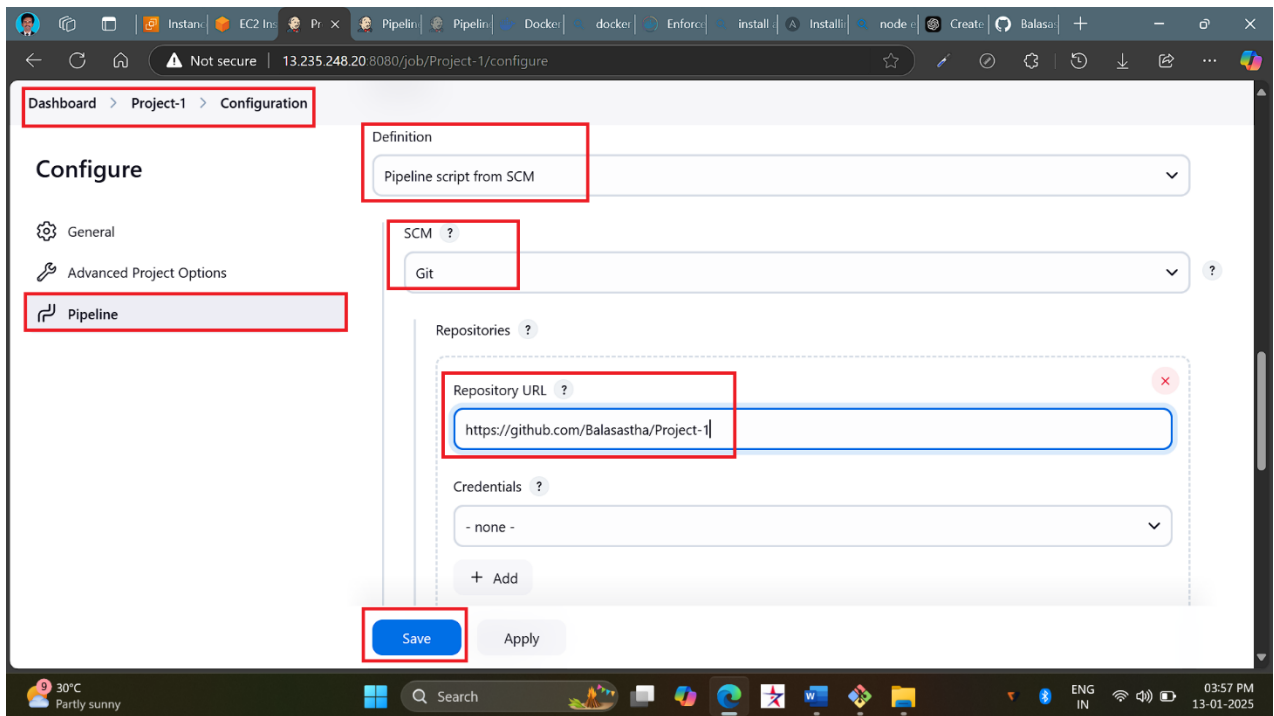
```
MINGW64/c/Users/Lenovo/Desktop/Project-1
pipeline {
  agent any
  stages {
    stage('Git checkout'){
      steps {
        {
          echo 'Checking out the git repo...'
          git url:' https://github.com/Balasastha/project-1.git'
        }
      }
    }
    stage('Package the code'){
      steps {
        {
          echo 'Packaging the code for you...'
          sh 'mvn clean package'
        }
      }
    }
    stage ('Build docker image'){
      steps {
        {
          echo 'Building docker image for you..'
          sh 'docker build -t balasastha/banking-finance:1 .'
        }
      }
    }
    stage ('Docker login'){
      steps {
        {
          echo 'Logging in into docker...'
          withCredentials([string(credentialsId: 'dockerhubpass', variable: 'dockerhubpass')]) {
            sh 'docker login -u balasastha -p ${dockerhubpass}'
          }
        }
      }
    }
    stage ('Push image into docker'){
      steps {
        {
          echo 'Pushing the docker image for you'
          sh 'docker push balasastha/banking-finance:1'
        }
      }
    }
    stage ('Invoking ansible playbook'){
      steps {
        {
          echo 'Running ansible playbook in test server...'
        }
      }
    }
  }
}
```

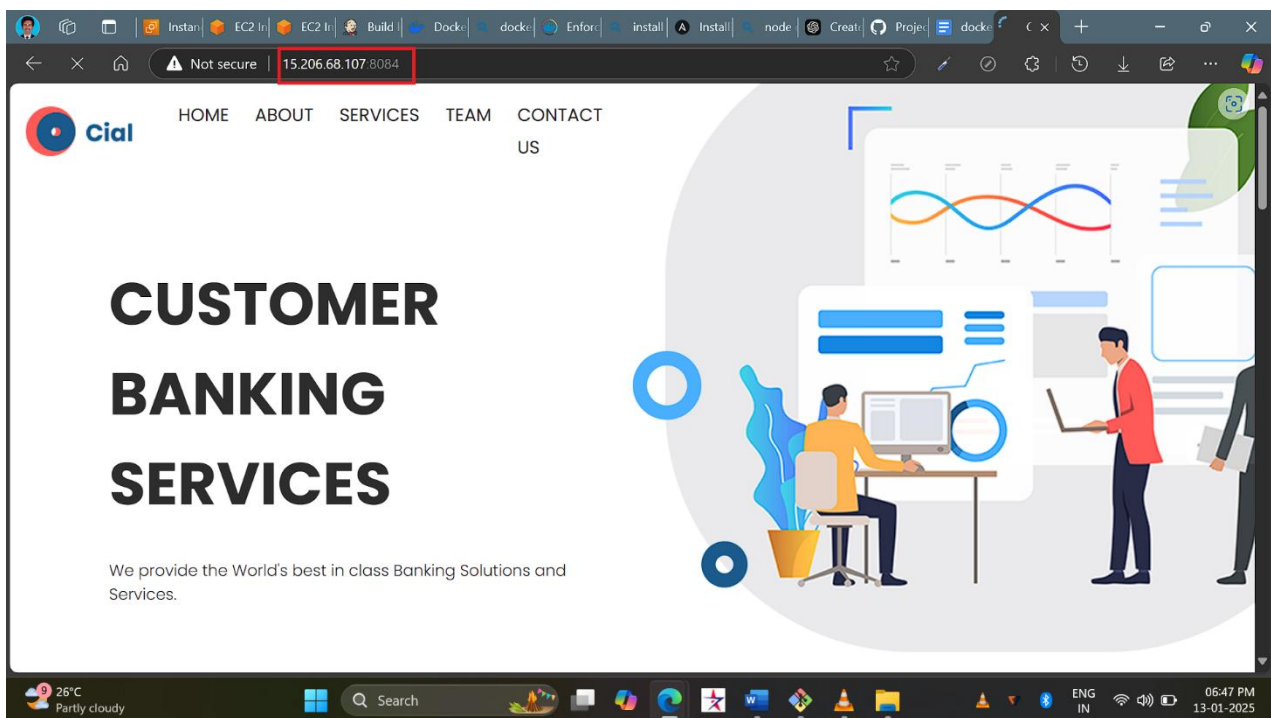
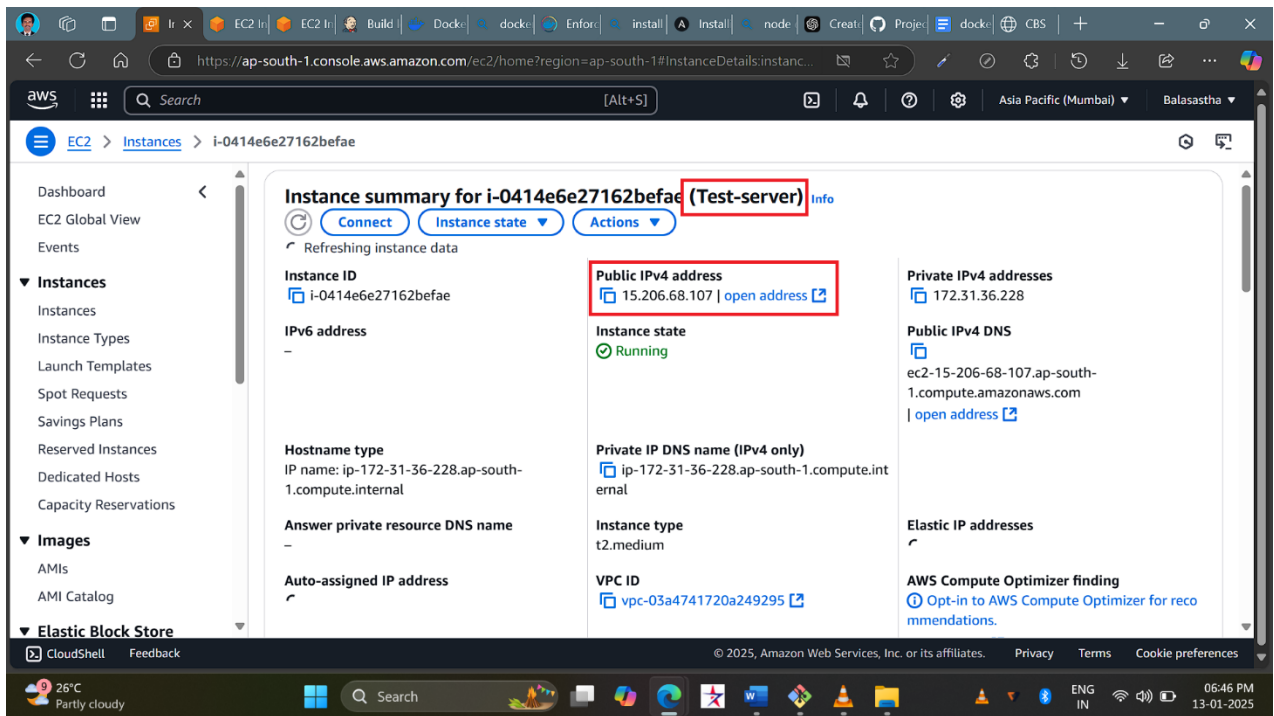
Jenkinsfile[+] [unix] (05:29 01/01/1970) 1,1 Top

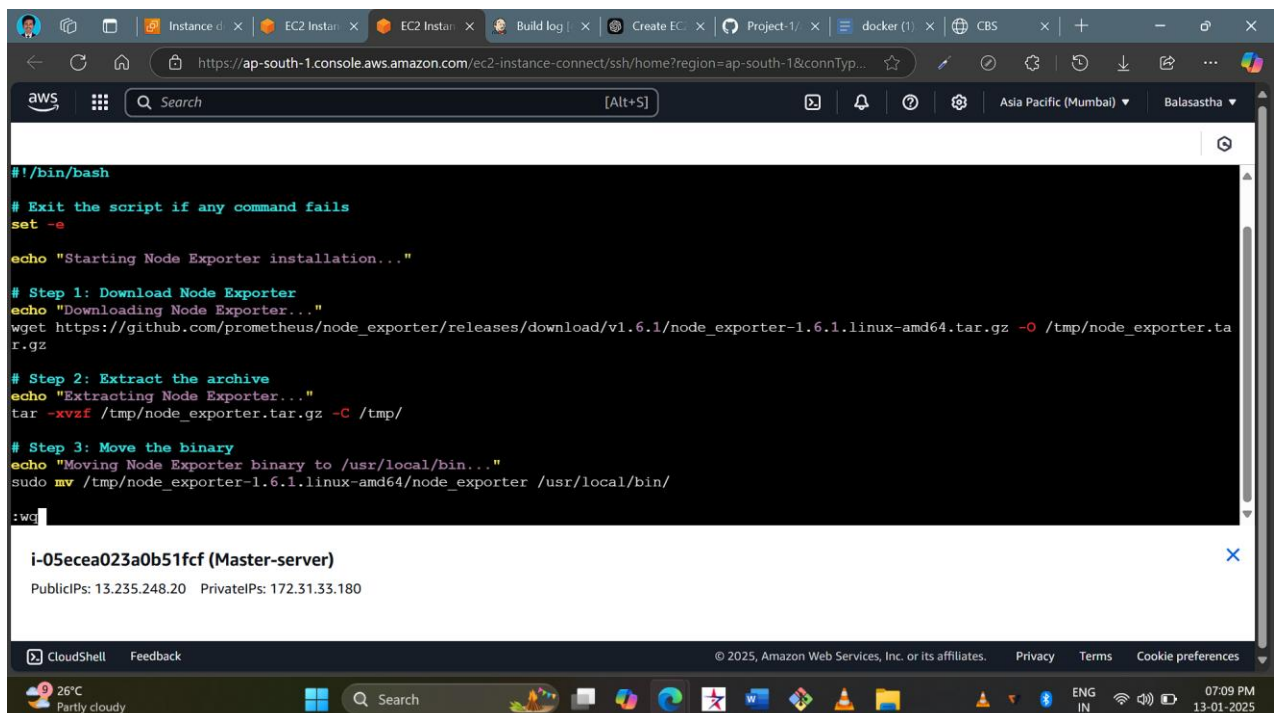
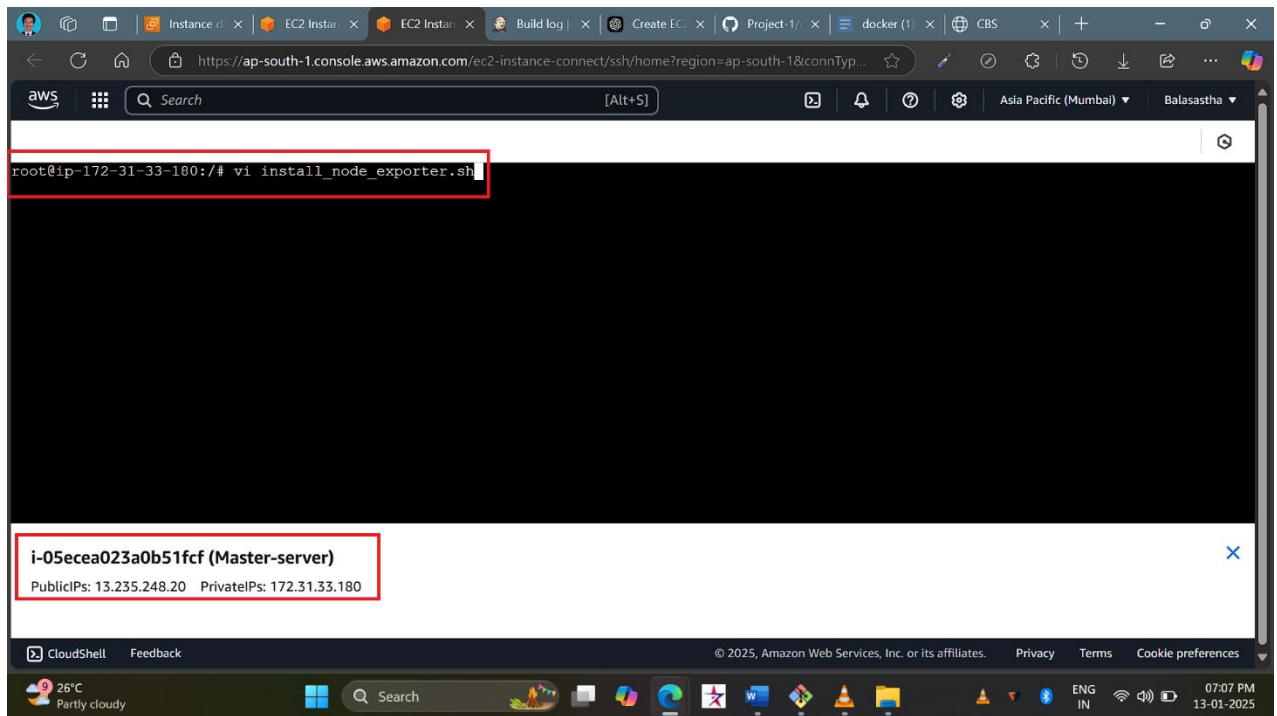
30°C Haze

```
MINGW64/c/Users/Lenovo/Desktop/Project-1
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ vi Jenkinsfile
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git add Jenkinsfile
warning: in the working copy of 'Jenkinsfile', LF will be replaced by CRLF the next time Git touches it
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git commit -m "Third commit"
[Master f2fc43a] Third commit
1 file changed, 52 insertions(+)
create mode 100644 Jenkinsfile
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git push origin -u master
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 753 bytes | 753.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Balasastha/Project-1
   35c8625..f2fc43a master -> master
branch 'master' set up to track 'origin/master'.
lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$
```

30°C Haze







Instance d... x EC2 Instar... x EC2 Instar... x Build log | x Create EC... x Project 1/... x docker (1) x CBS x + - @ x

https://ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connTyp...

aws Search [Alt+S] Asia Pacific (Mumbai) Balasastha

```
root@ip-172-31-33-180:~# vi install_node_exporter.sh
root@ip-172-31-33-180:~# chmod +x install_node_exporter.sh
root@ip-172-31-33-180:~# ./install_node_exporter.sh
Starting Node Exporter installation...
Downloading Node Exporter...
--2025-01-13 13:40:01-- https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response...
```

i-05ecea023a0b51fcf (Master-server)
PublicIPs: 13.235.248.20 PrivateIPs: 172.31.33.180

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26°C Mostly cloudy Search 07:10 PM 13-01-2025

Instance d... x EC2 Instar... x EC2 Instar... x Build log | x Create EC... x Project 1/... x docker (1) x CBS x + - @ x

https://ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connTyp...

aws Search [Alt+S] Asia Pacific (Mumbai) Balasastha

```
root@ip-172-31-36-228:/home/ubuntu# vi install_node_exporter.sh
```

i-0414e6e27162befae (Test-server)
PublicIPs: 15.206.68.107 PrivateIPs: 172.31.36.228

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26°C Mostly cloudy Search 07:11 PM 13-01-2025

```
#!/bin/bash

# Exit the script if any command fails
set -e

echo "Starting Node Exporter installation..."

# Step 1: Download Node Exporter
echo "Downloading Node Exporter..."
wget https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz -O /tmp/node_exporter.tar.gz

# Step 2: Extract the archive
echo "Extracting Node Exporter..."
tar -xvzf /tmp/node_exporter.tar.gz -C /tmp/

# Step 3: Move the binary
echo "Moving Node Exporter binary to /usr/local/bin..."
sudo mv /tmp/node_exporter-1.6.1.linux-amd64/node_exporter /usr/local/bin/

:wg
```

i-0414e6e27162befae (Test-server)
PublicIPs: 15.206.68.107 PrivateIPs: 172.31.36.228

```
root@ip-172-31-36-228:/home/ubuntu# vi install_node_exporter.sh
root@ip-172-31-36-228:/home/ubuntu# chmod +x install_node_exporter.sh
root@ip-172-31-36-228:/home/ubuntu# ./install_node_exporter.sh
Starting Node Exporter installation...
Downloading Node Exporter...
--2025-01-13 13:42:59-- https://github.com/prometheus/node_exporter/releases/download/v1.6.1/node_exporter-1.6.1.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response...
```

i-0414e6e27162befae (Test-server)
PublicIPs: 15.206.68.107 PrivateIPs: 172.31.36.228

Instan: X EC2 I: X EC2 I: X Build: X Projec: X dock: X Creat: X CBS: X 13.23: X 15.20: X + -

Not secure | 13.235.248.20:9100/metrics

```
# HELP go_gc_duration_seconds A summary of the pause duration of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 8
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.20.6"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated and still in use.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 948384
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated, even if freed.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 948384
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.44548e+06
# HELP go_memstats_frees_total Total number of frees.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 682
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 7.196104e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and still in use.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 948384
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used.
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 1.458176e+06
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use.
# TYPE go_memstats_heap_inuse_bytes gauge
```

26°C Mostly cloudy Search 07:15 PM 13-01-2025

Instan: X EC2 I: X EC2 I: X Build: X Projec: X dock: X Creat: X CBS: X 13.23: X 15.20: X + -

Not secure | 15.206.68.107:9100/metrics

```
# HELP go_gc_duration_seconds A summary of the pause duration of garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
# HELP go_goroutines Number of goroutines that currently exist.
# TYPE go_goroutines gauge
go_goroutines 7
# HELP go_info Information about the Go environment.
# TYPE go_info gauge
go_info{version="go1.20.6"} 1
# HELP go_memstats_alloc_bytes Number of bytes allocated and still in use.
# TYPE go_memstats_alloc_bytes gauge
go_memstats_alloc_bytes 930016
# HELP go_memstats_alloc_bytes_total Total number of bytes allocated, even if freed.
# TYPE go_memstats_alloc_bytes_total counter
go_memstats_alloc_bytes_total 930016
# HELP go_memstats_buck_hash_sys_bytes Number of bytes used by the profiling bucket hash table.
# TYPE go_memstats_buck_hash_sys_bytes gauge
go_memstats_buck_hash_sys_bytes 1.445336e+06
# HELP go_memstats_frees_total Total number of frees.
# TYPE go_memstats_frees_total counter
go_memstats_frees_total 729
# HELP go_memstats_gc_sys_bytes Number of bytes used for garbage collection system metadata.
# TYPE go_memstats_gc_sys_bytes gauge
go_memstats_gc_sys_bytes 7.1776e+06
# HELP go_memstats_heap_alloc_bytes Number of heap bytes allocated and still in use.
# TYPE go_memstats_heap_alloc_bytes gauge
go_memstats_heap_alloc_bytes 930016
# HELP go_memstats_heap_idle_bytes Number of heap bytes waiting to be used.
# TYPE go_memstats_heap_idle_bytes gauge
go_memstats_heap_idle_bytes 1.622016e+06
# HELP go_memstats_heap_inuse_bytes Number of heap bytes that are in use.
# TYPE go_memstats_heap_inuse_bytes gauge
```

26°C Mostly cloudy Search 07:15 PM 13-01-2025

aws console window showing terminal output for Ubuntu system updates and a terminal window showing the command `sudo su` and the output `root@ip-172-31-41-38:/home/ubuntu#`. A red box highlights the command `!- setup_prometheus_and_node_exporter.sh`.

PublicIPs: 52.66.194.245 PrivateIPs: 172.31.41.38

aws console window showing terminal output for Prometheus and Node Exporter setup. The terminal output includes the following commands and their outputs:

```
#!/bin/bash
# Exit script on any error
set -e

echo "Starting setup for Prometheus and Node Exporter..."

# --- Install Prometheus ---
echo "Downloading Prometheus..."
wget https://github.com/prometheus/prometheus/releases/download/v2.47.0/prometheus-2.47.0.linux-amd64.tar.gz -O /tmp/prometheus.tar.gz

echo "Extracting Prometheus..."
tar xvfz /tmp/prometheus.tar.gz -C /tmp/

echo "Moving Prometheus binaries to /usr/local/bin..."
sudo mv /tmp/prometheus-2.47.0.linux-amd64/prometheus /usr/local/bin/
sudo mv /tmp/prometheus-2.47.0.linux-amd64/promtool /usr/local/bin/

echo "Moving Prometheus configuration and data directories..."
sudo mkdir -p /etc/prometheus /var/lib/prometheus
```

PublicIPs: 52.66.194.245 PrivateIPs: 172.31.41.38

The programs included with the Ubuntu system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law.

To run a command as administrator (user "root"), use "sudo <command>". See "man sudo_root" for details.

```
ubuntu@ip-172-31-41-38:~$ sudo su
root@ip-172-31-41-38:/home/ubuntu# vi setup_prometheus_and_node_exporter.sh
root@ip-172-31-41-38:/home/ubuntu# chmod +x setup_prometheus_and_node_exporter.sh
root@ip-172-31-41-38:/home/ubuntu# ./setup_prometheus_and_node_exporter.sh
Starting setup for Prometheus and Node Exporter...
Downloading Prometheus...
--2025-01-13 14:06:42-- https://github.com/prometheus/prometheus/releases/download/v2.47.0/prometheus-2.47.0.linux-amd64.tar.gz
Resolving github.com (github.com)... 20.207.73.82
Connecting to github.com (github.com)|20.207.73.82|:443... connected.
HTTP request sent, awaiting response...
```

i-0958f51b452a8a2fa (Grafana)
PublicIPs: 52.66.194.245 PrivateIPs: 172.31.41.38

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26°C Mostly cloudy 07:36 PM 13-01-2025

Not secure | 52.66.194.245 9090/graph?g0.expr=&g0.tab=1&g0.stacked=0&g0.show_exemplars=0&g0.ra...

Prometheus Alerts Graph Status Help

☐ Use local time ☐ Enable query history ☒ Enable autocomplete ☒ Enable highlighting ☒ Enable linter

Q Expression (press Shift+Enter for newlines) Execute

Table Graph

< Evaluation time >

No data queried yet

Remove Panel

Add Panel

26°C Mostly cloudy 07:38 PM 13-01-2025

Browser window showing the Prometheus Targets page. The URL is `52.66.194.245:9090/targets?search=`. The page displays a table of targets for the `node_exporter` job.

Targets

All scrape pools All Unhealthy Collapse All Unknown Unhealthy Healthy

node_exporter (2/2 up) [show less](#)

Endpoint	State	Labels	Last Scrape	Scrape Duration	Error
http://13.235.248.20:9100/metrics	UP	<code>instance="13.235.248.20:9100"</code> <code>job="node_exporter"</code>	2.881s ago	12.595ms	
http://15.206.68.107:9100/metrics	UP	<code>instance="15.206.68.107:9100"</code> <code>job="node_exporter"</code>	1.695s ago	14.613ms	

System tray: 26°C Mostly cloudy, Search, 07:54 PM 13-01-2025

Browser window showing the AWS Management Console. The URL is `https://ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?region=ap-south-1&connTyp...`. The page displays a terminal window with the command `vi install_grafana.sh` entered.

aws [Alt+S] Asia Pacific (Mumbai) Balasastha

root@ip-172-31-41-38:/home/ubuntu# `vi install_grafana.sh`

i-0958f51b452a8a2fa (Grafana)
PublicIPs: 52.66.194.245 PrivateIPs: 172.31.41.38

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System tray: 26°C Mostly cloudy, Search, 07:50 PM 13-01-2025

```
#!/bin/bash

# Update the package list
sudo apt-get update -y

# Install required software properties package
sudo apt-get install -y software-properties-common

# Add the Grafana repository
sudo add-apt-repository "deb [arch=amd64] https://packages.grafana.com/oss/deb stable main"

# Add Grafana GPG key
wget -q -O - https://packages.grafana.com/gpg.key | sudo apt-key add -

# Update the package list after adding the Grafana repository
sudo apt-get update -y

# Install Grafana
sudo apt-get install -y grafana

:wq
```

i-0958f51b452a8a2fa (Grafana)
PublicIPs: 52.66.194.245 PrivateIPs: 172.31.41.38

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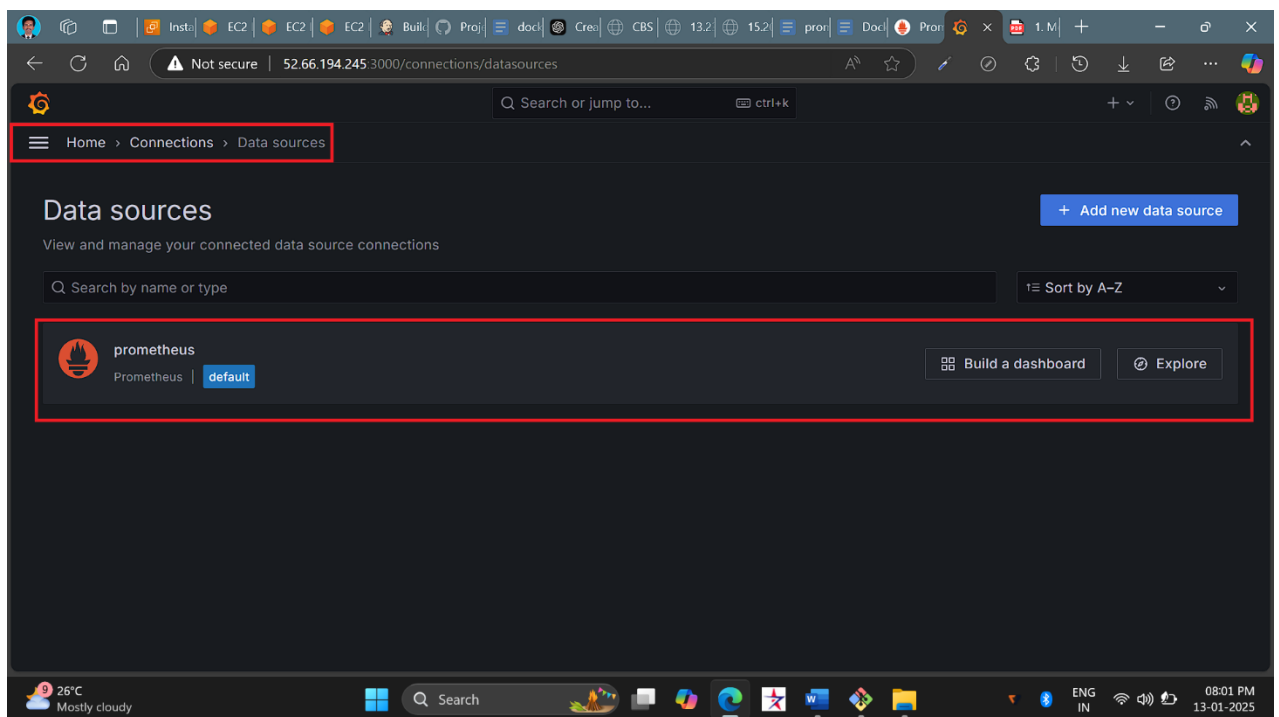
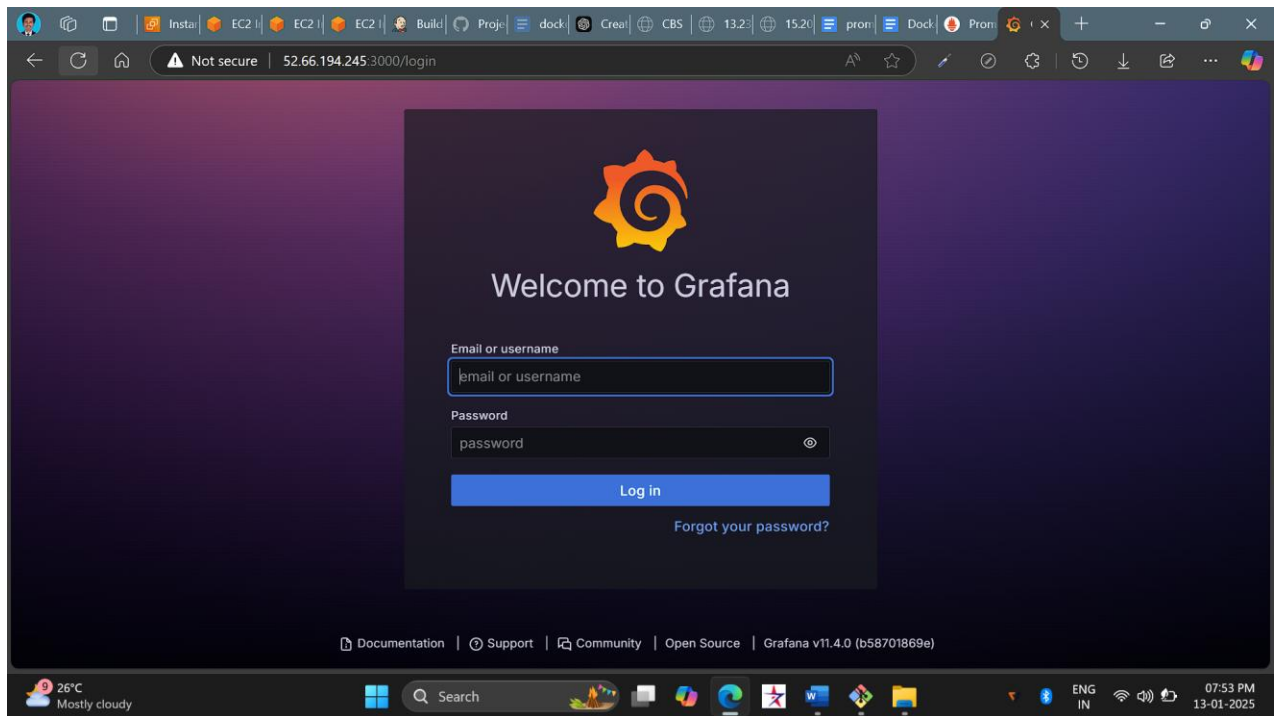
26°C Mostly cloudy 07:51 PM 13-01-2025

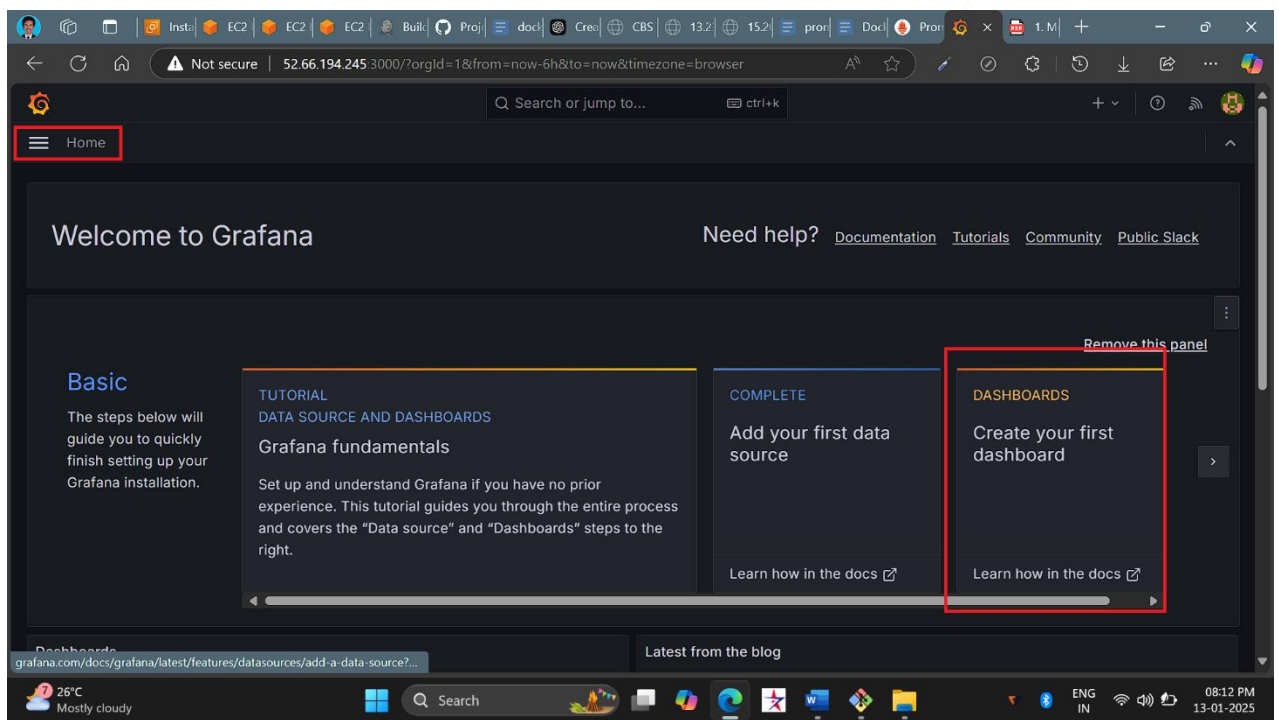
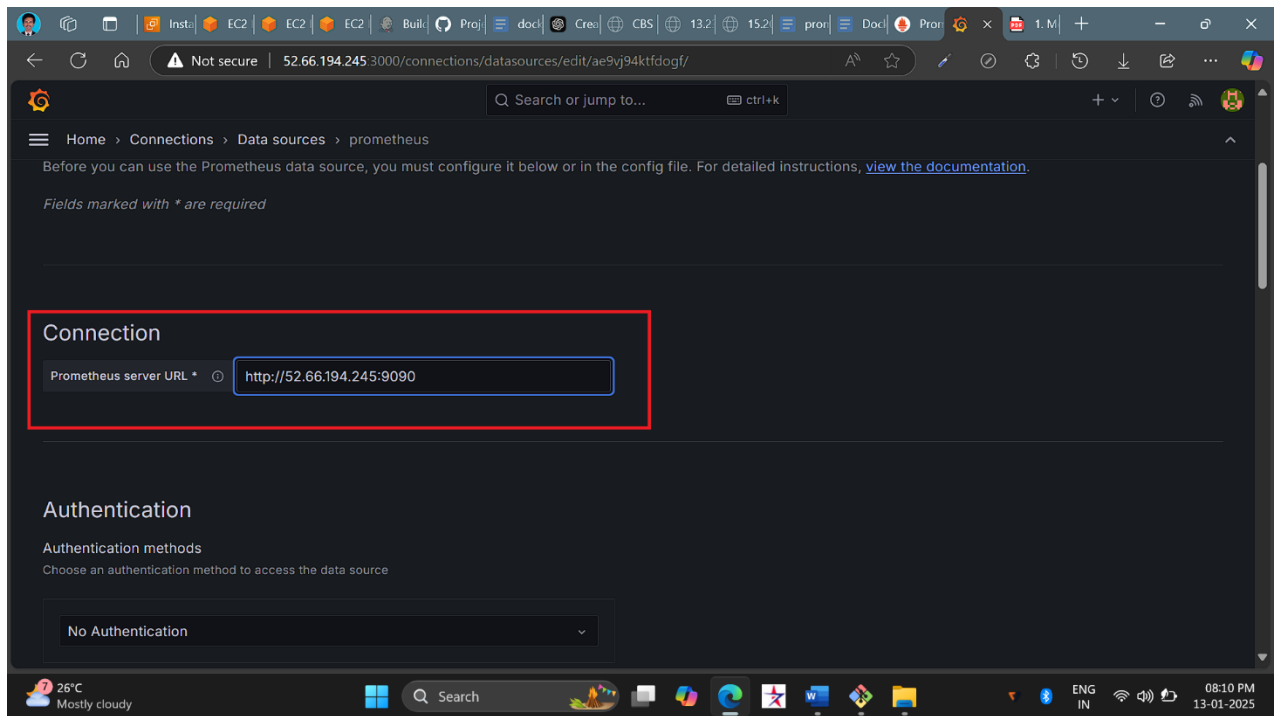
```
root@ip-172-31-41-38:/home/ubuntu# vi install_grafana.sh
root@ip-172-31-41-38:/home/ubuntu# chmod +x install_grafana.sh
root@ip-172-31-41-38:/home/ubuntu# ./install_grafana.sh
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [151 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [310 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:8 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [208 B]
0% [4 Components-amd64 store 0 B] [8 InRelease 14.2 kB/126 kB 11%]
```

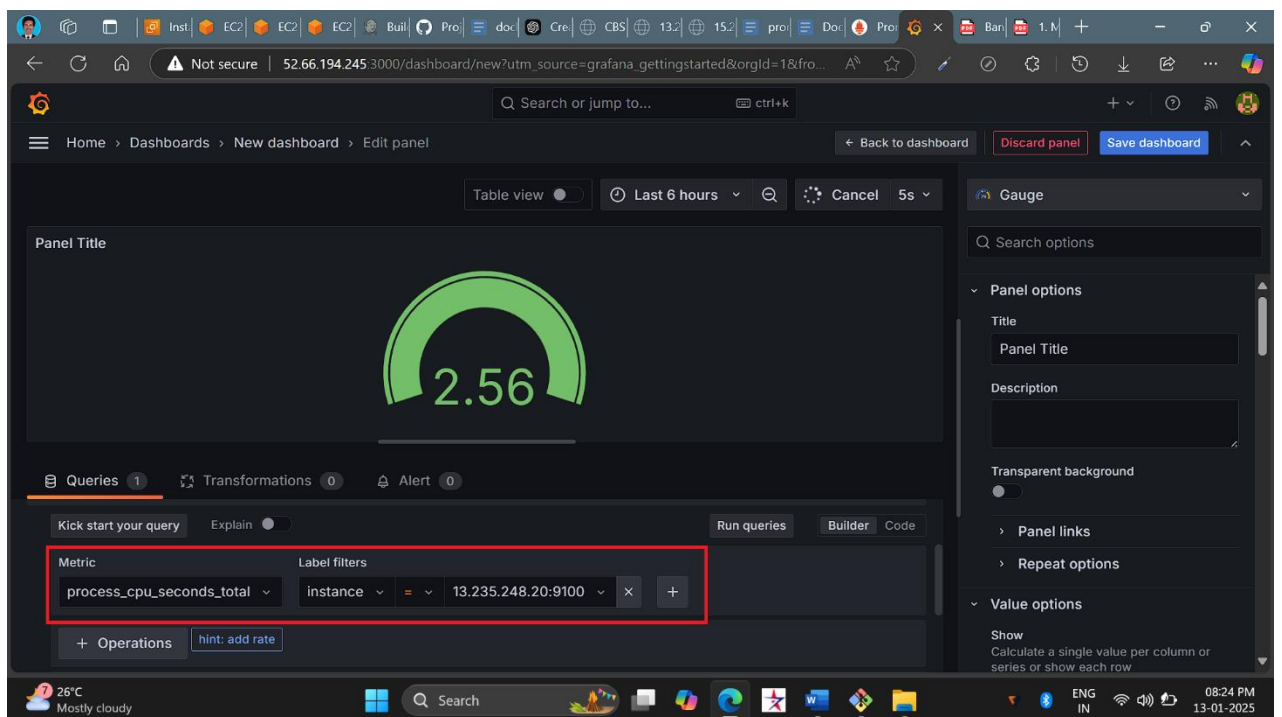
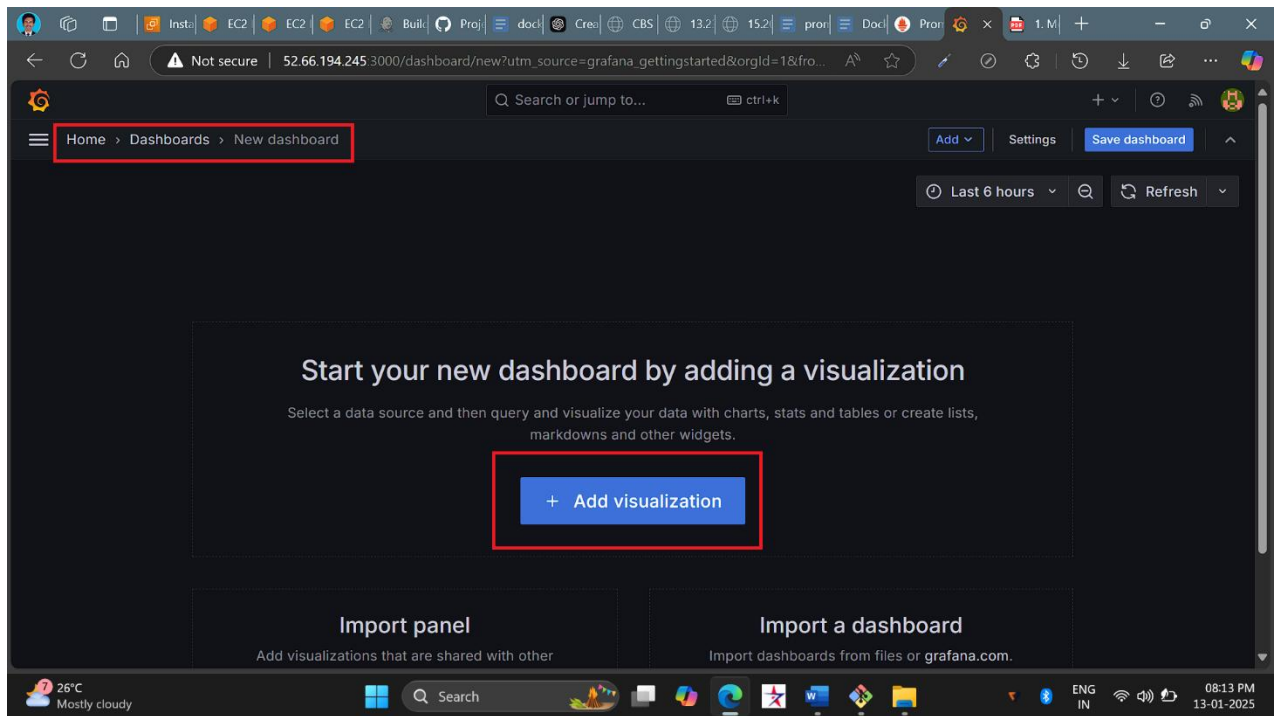
i-0958f51b452a8a2fa (Grafana)
PublicIPs: 52.66.194.245 PrivateIPs: 172.31.41.38

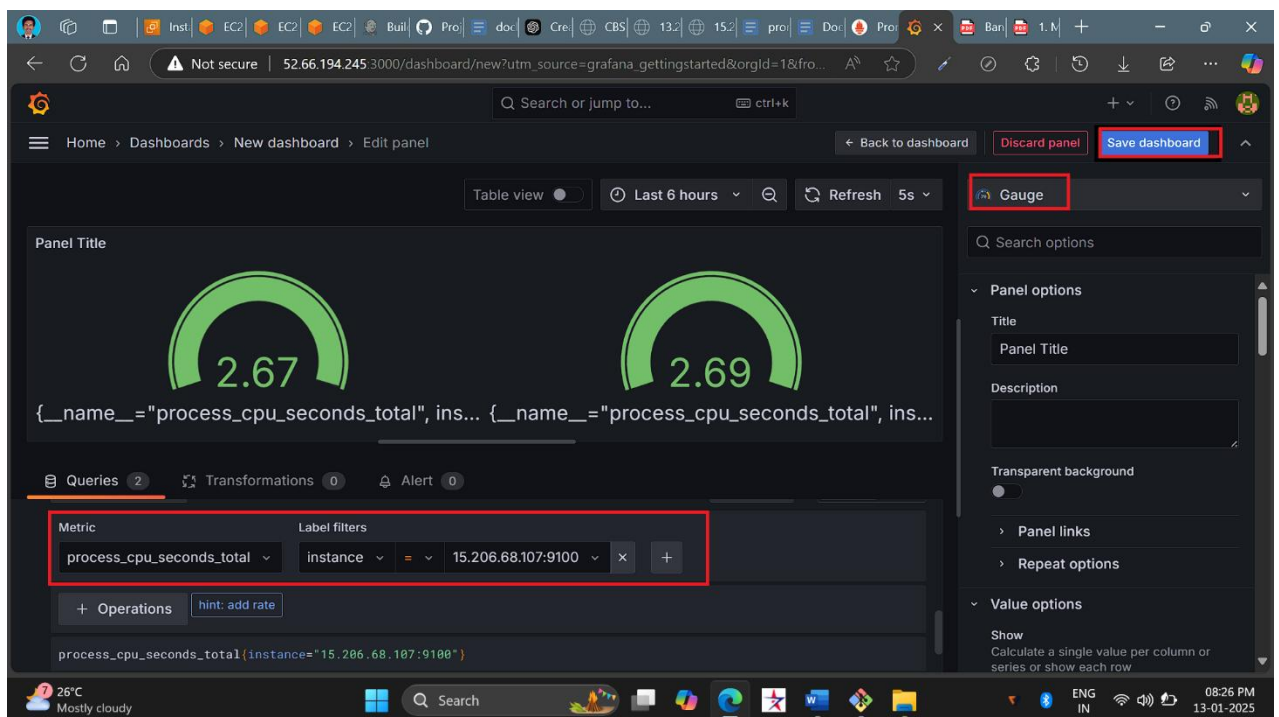
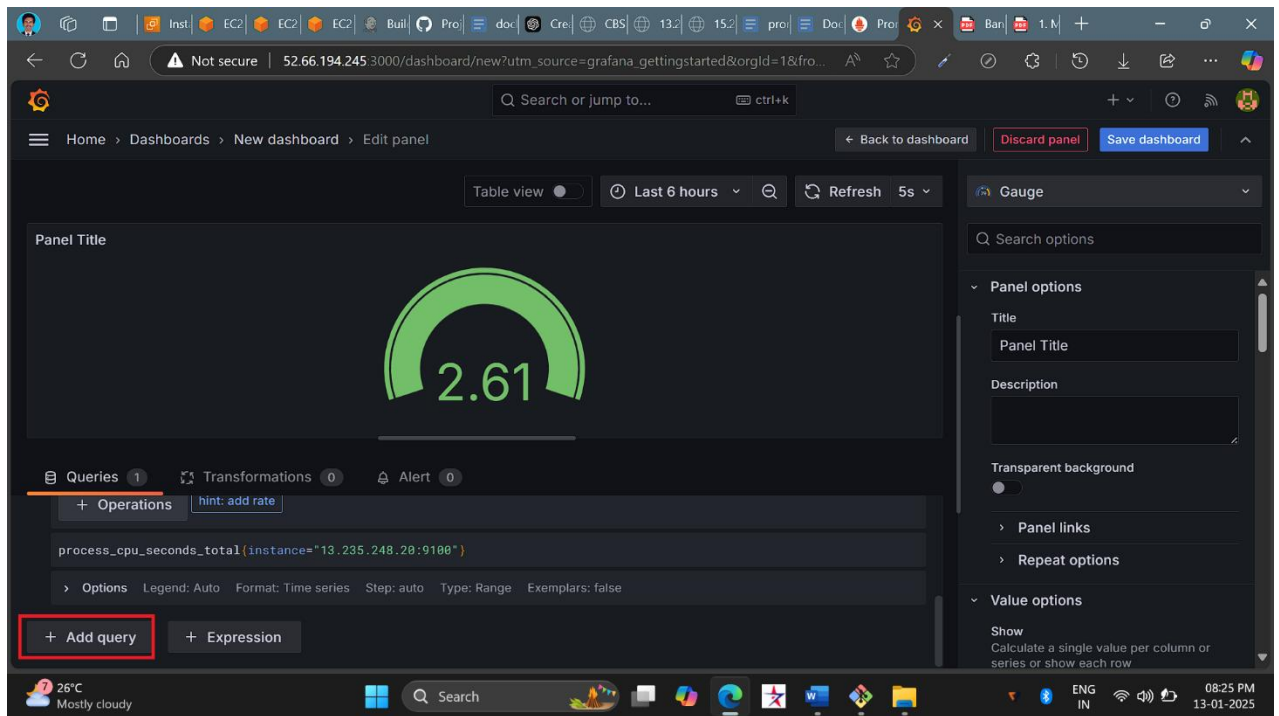
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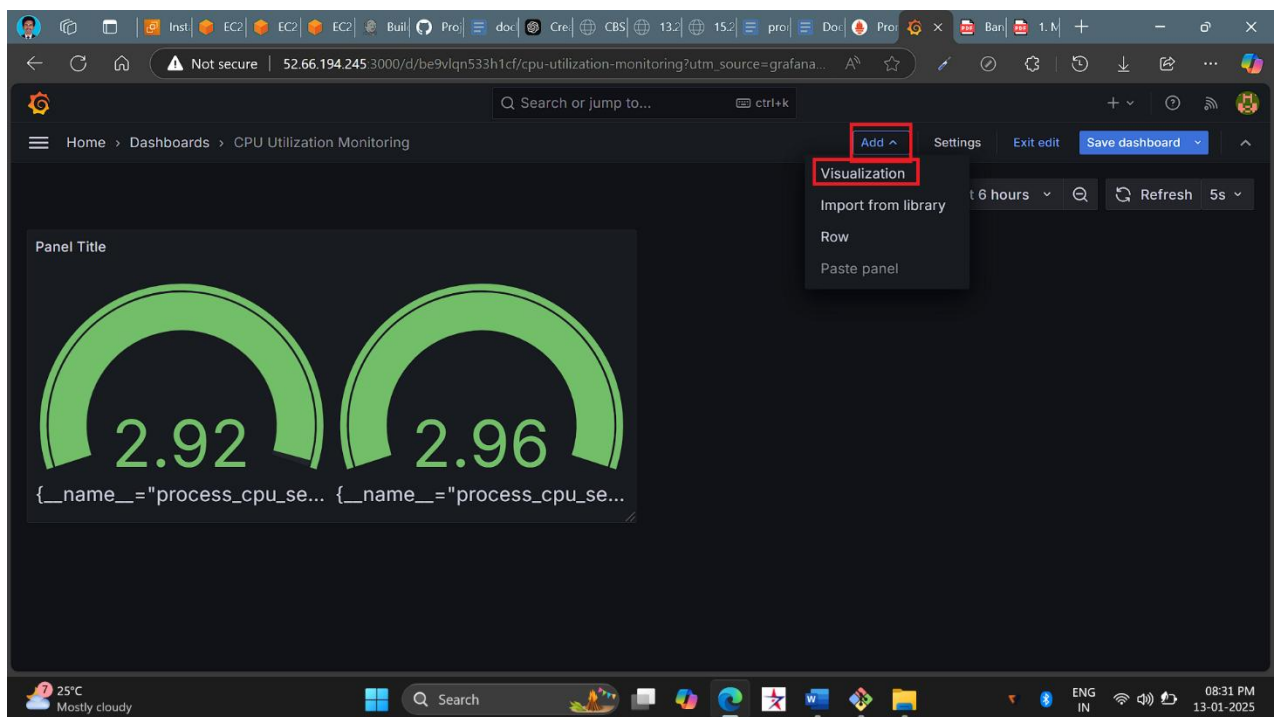
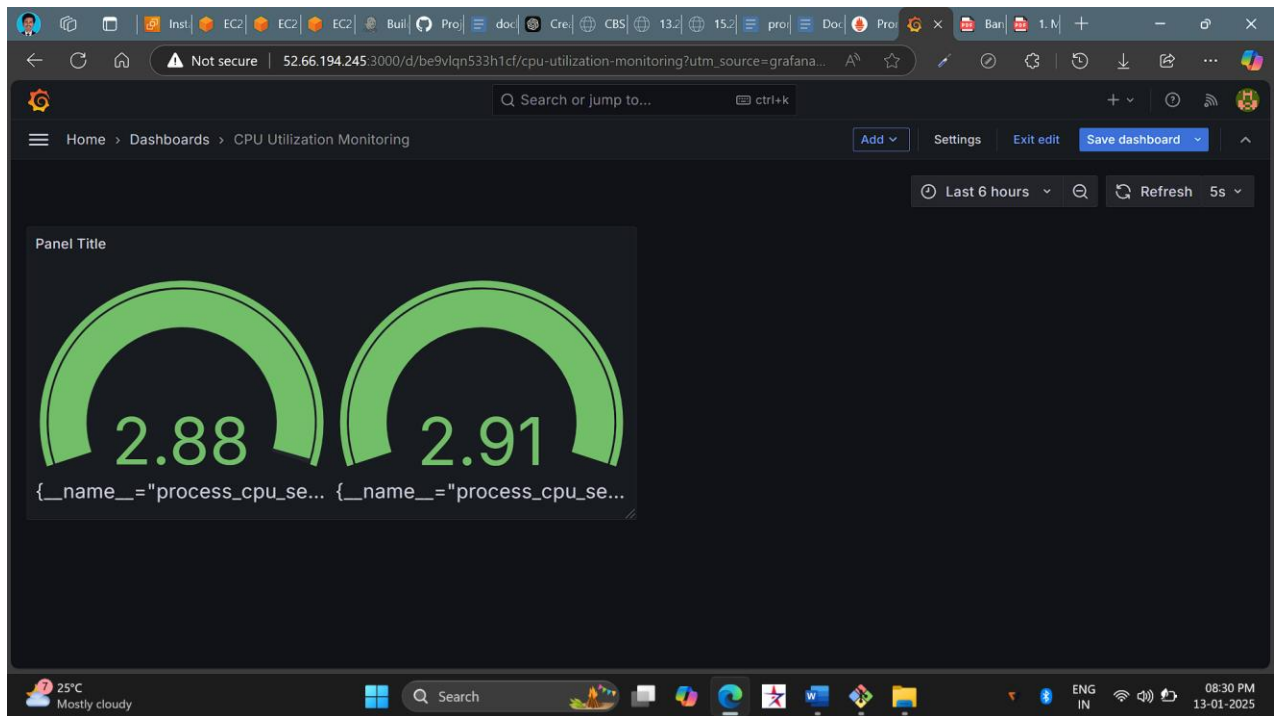
26°C Mostly cloudy 07:51 PM 13-01-2025

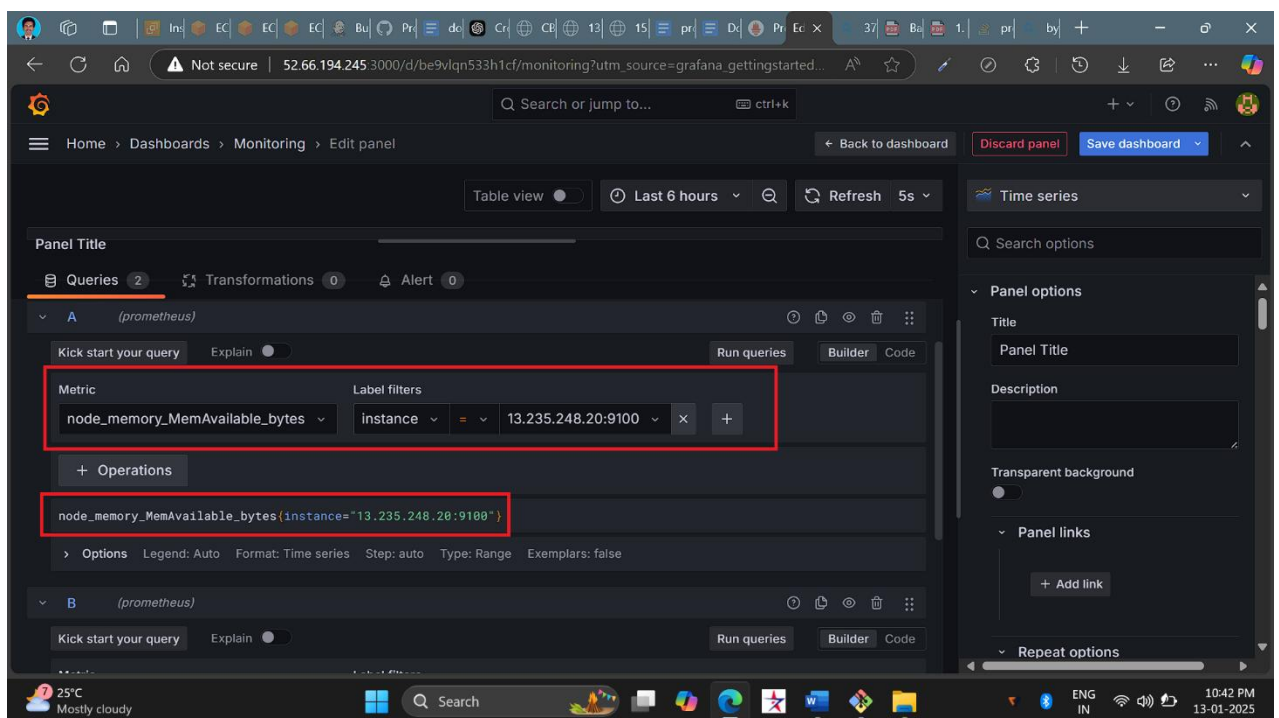
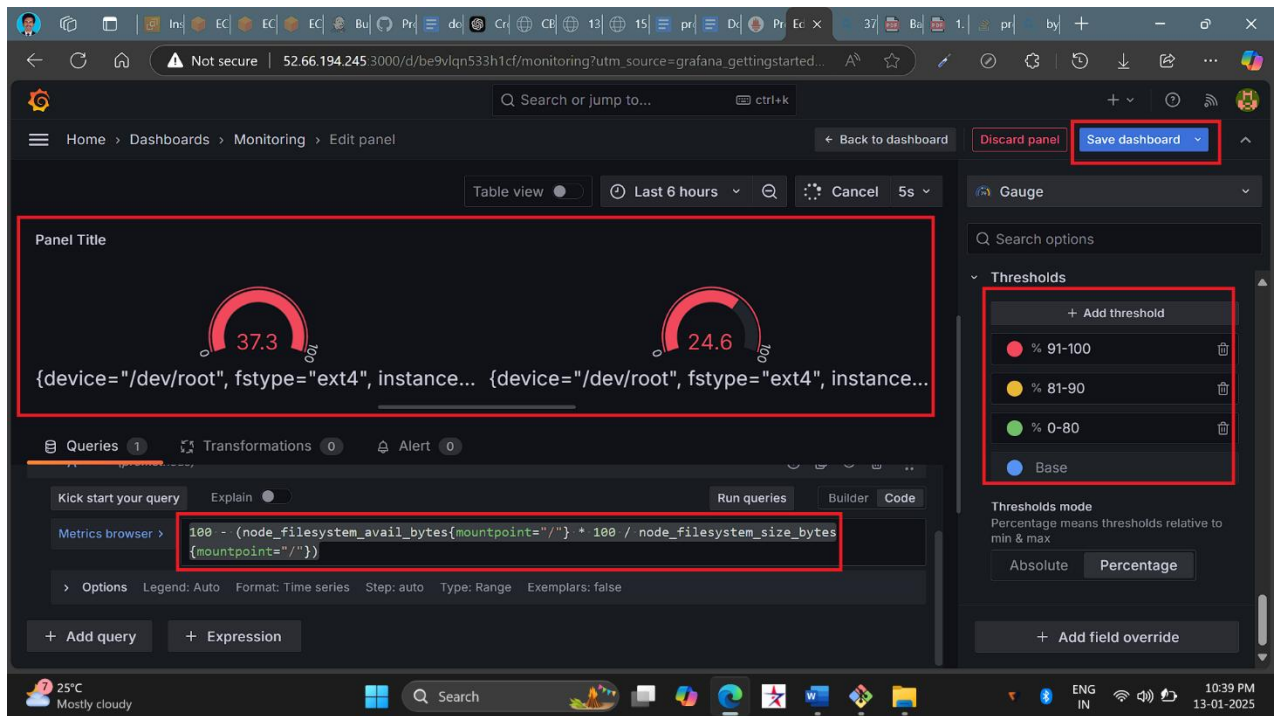












Panel Title

Queries 2 Transformations 0 Alert 0

Options Legend: Auto Format: Time series Step: auto Type: Range Exemplars: false

B (prometheus)

Kick start your query Explain Run queries Builder Code

Metric: node_memory_MemAvailable_bytes Label filters: instance = 15.206.68.107:9100

+ Operations

node_memory_MemAvailable_bytes(instance="15.206.68.107:9100")

Options Legend: Auto Format: Time series Step: auto Type: Range Exemplars: false

+ Add query + Expression

Time series

Search options

Panel options

Title: Panel Title

Description

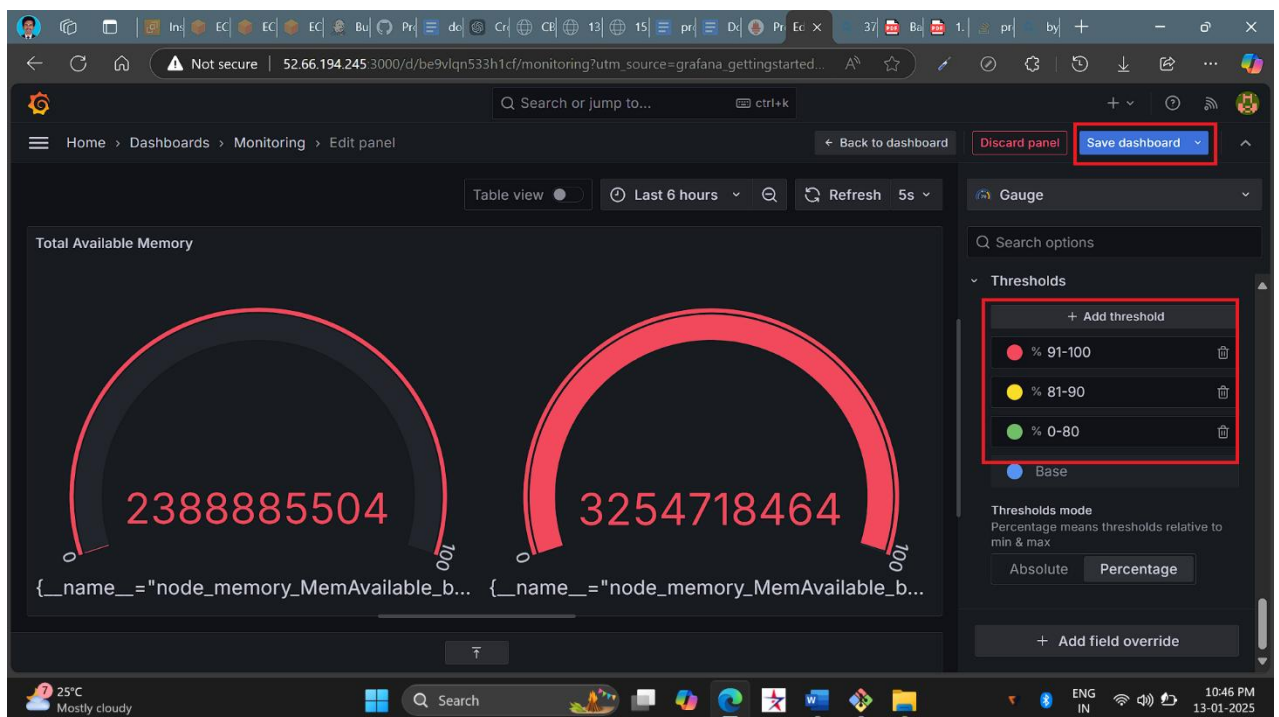
Transparent background

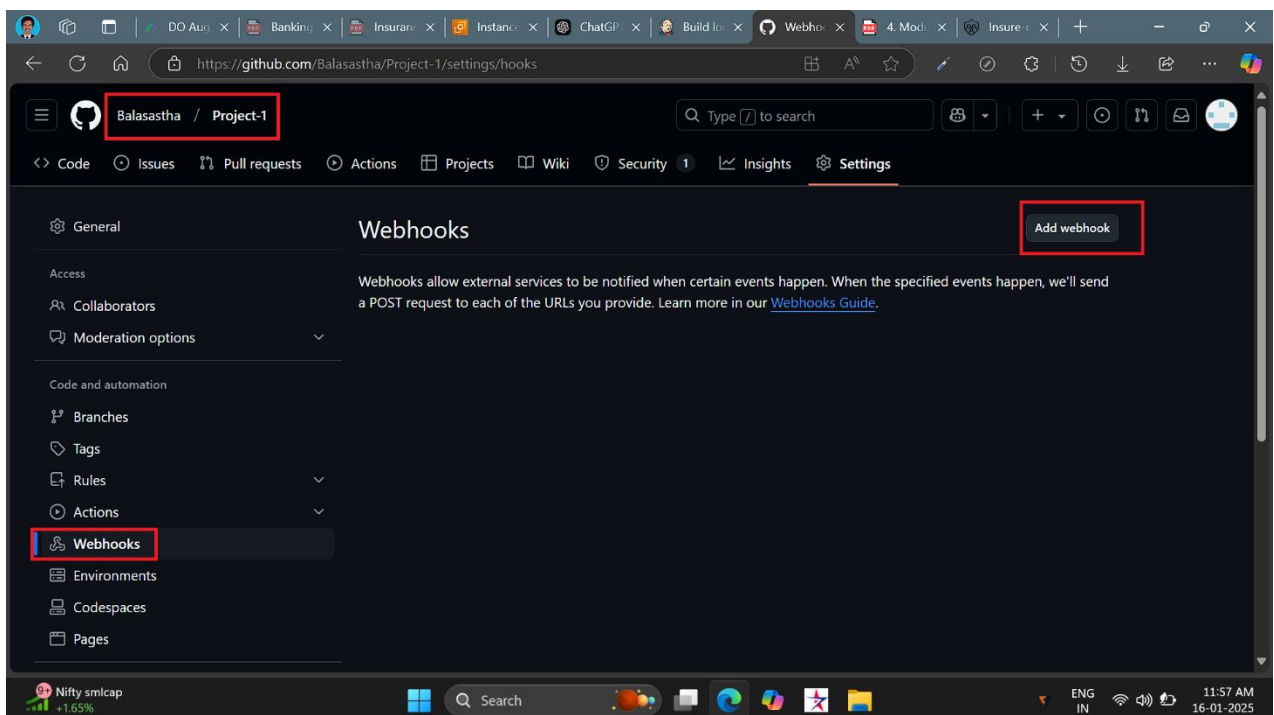
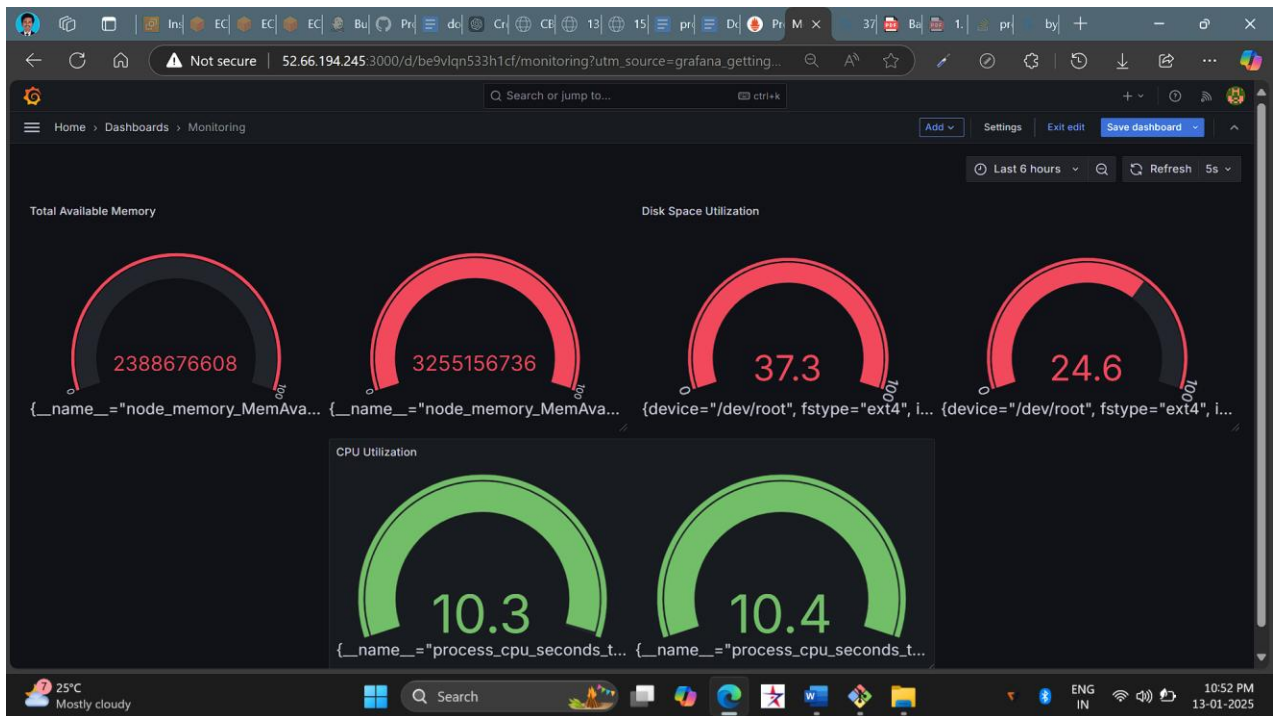
Panel links

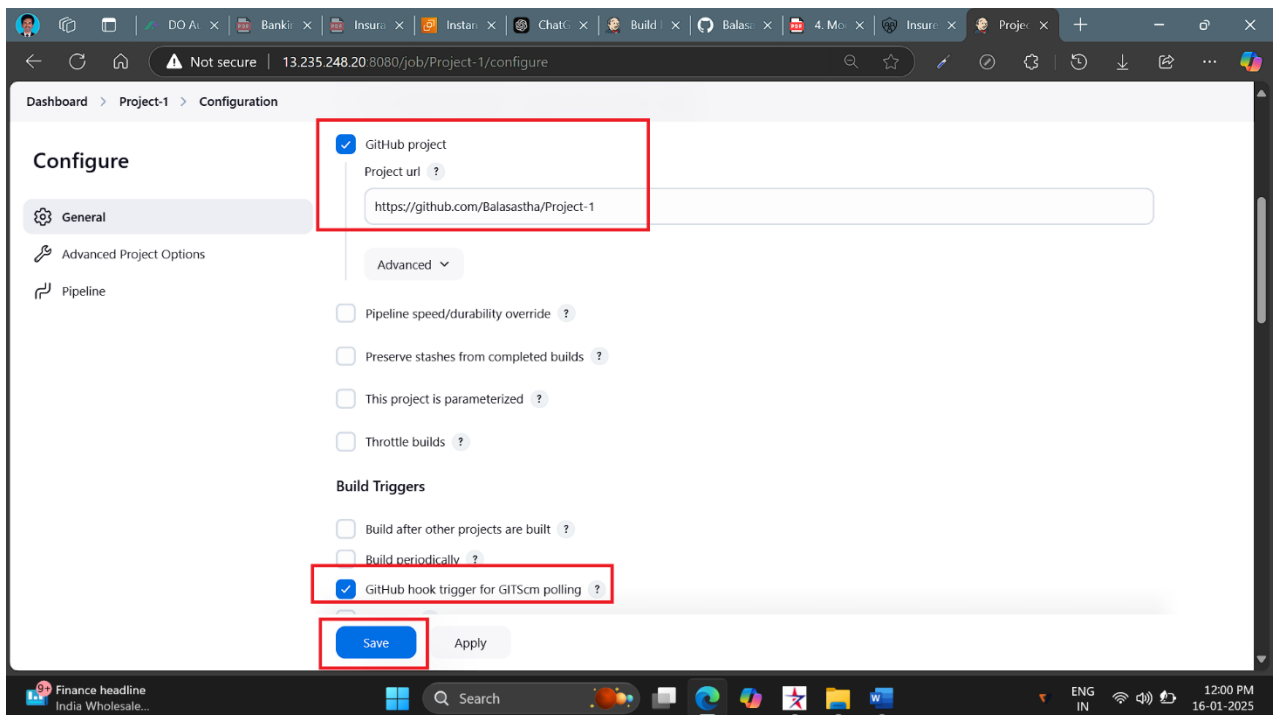
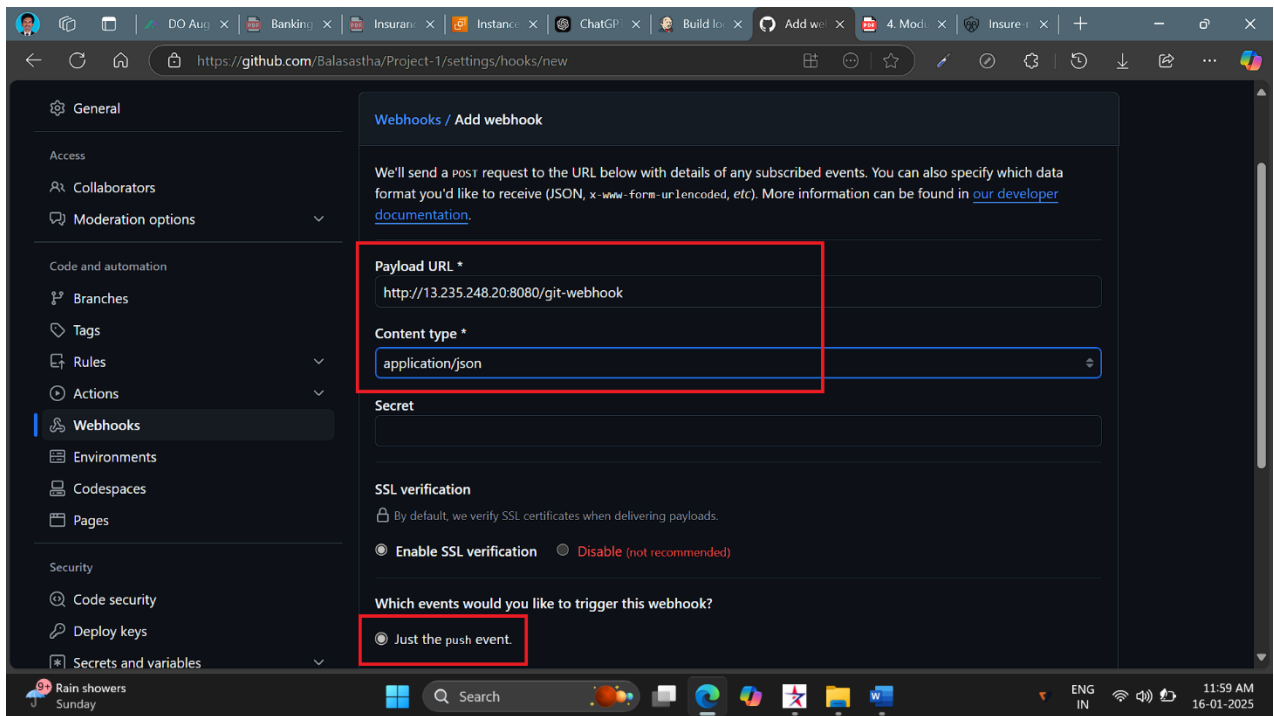
+ Add link

Repeat options

25°C Mostly cloudy 10:42 PM 13-01-2025







```
MINGW64/c/Users/Lenovo/Desktop/Project-1
vi readme.txt
```

```
MINGW64/c/Users/Lenovo/Desktop/Project-1
H1 welcome to banking and finance project

readme.txt[+] [unix] (05:29 01/01/1970) 1,42 All
-- INSERT --
```

```
MINGW64~/Users/Lenovo/Desktop/Project-1
Lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ vi readme.txt
Lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git add readme.txt
warning: in the working copy of 'readme.txt', LF will be replaced by CRLF the next time Git touches it
Lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git commit -m "Seventh commit"
[master cb27449] Seventh commit
1 file changed, 2 insertions(+)
create mode 100644 readme.txt
Lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ git push origin -u master
Counting objects: 100% (4/4), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 319 bytes | 319.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/Balasastha/Project-1
  2afe87e..cb27449 master -> master
branch 'master' set up to track 'origin/master'.
Lenovo@Balasastha MINGW64 ~/Desktop/Project-1 (master)
$ |
```

Dashboard > Project-2 >

Build Now

Configure

Delete Pipeline

GitHub

Stages

Rename

Pipeline Syntax

GitHub Hook Log

Builds

Filter

Today

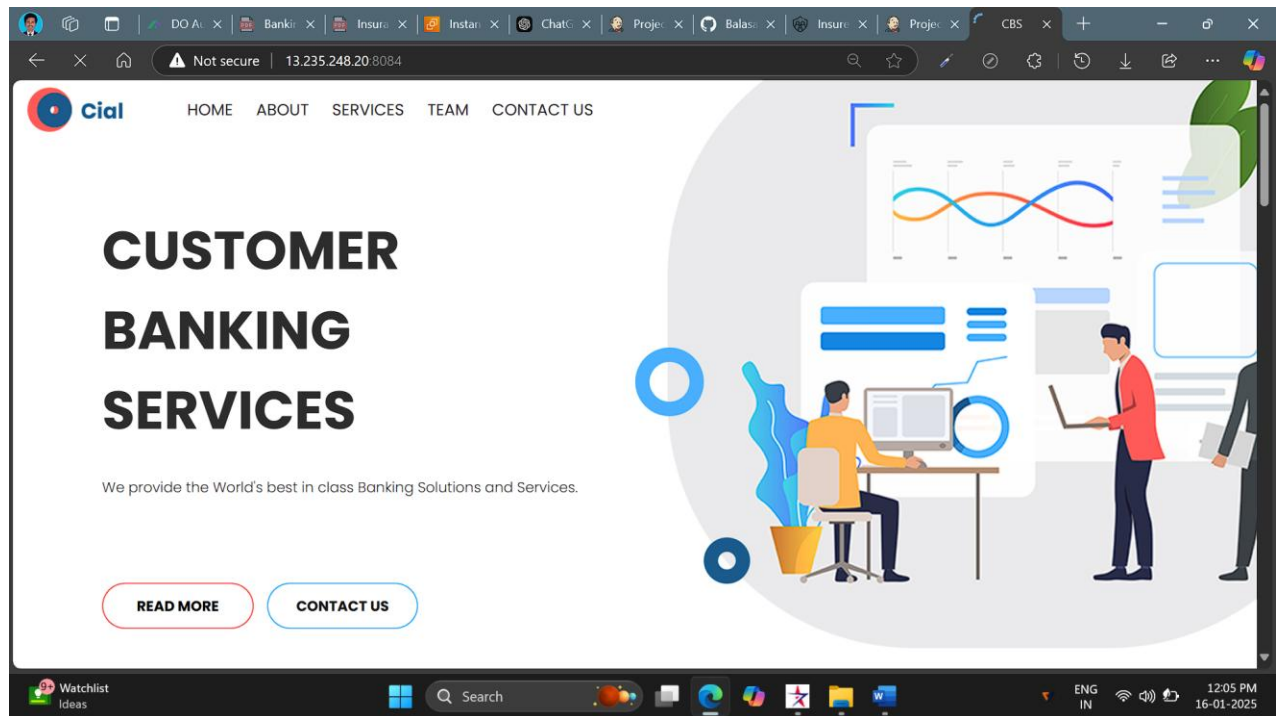
#5 6:35 AM

January 14, 2025

#1 1:03 PM

Permalinks

- Last build (#1), 1 day 17 hr ago
- Last stable build (#1), 1 day 17 hr ago
- Last successful build (#1), 1 day 17 hr ago
- Last completed build (#1), 1 day 17 hr ago



**BANKING AND FINANCE DOMAIN PROJECT
HAS BEEN SUCCESSFULLY COMPLETED
THANK YOU!!!**